

Elektrizität und Magnetismus

Kapitel 3: Elektrische Felder

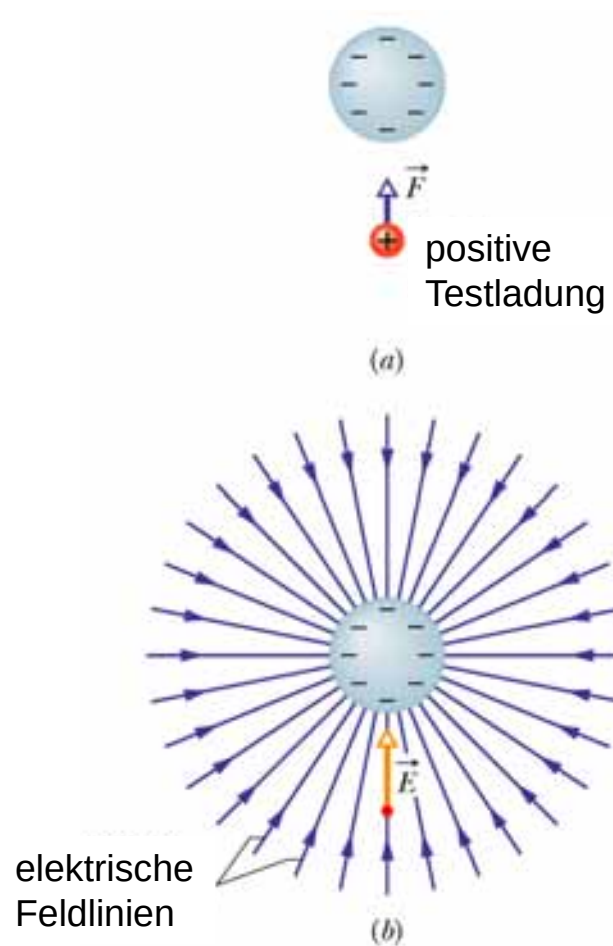
Prof. Axel Görlitz
Heinrich-Heine Universität Düsseldorf
Axel.Goerlitz@uni-duesseldorf.de

zu 3: Elektrische Felder



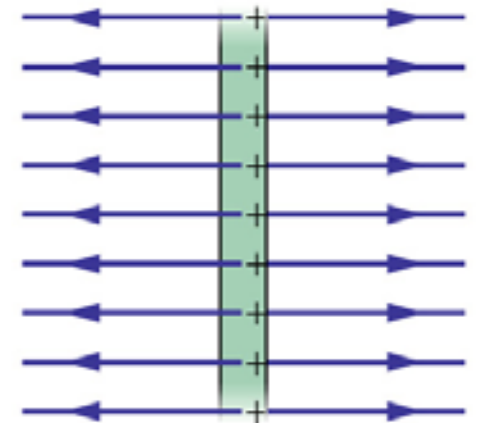
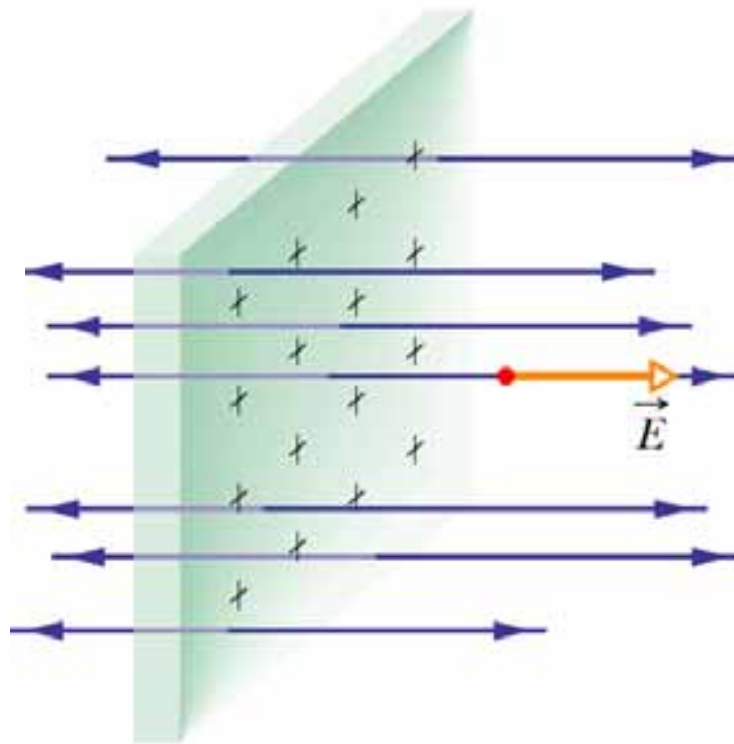
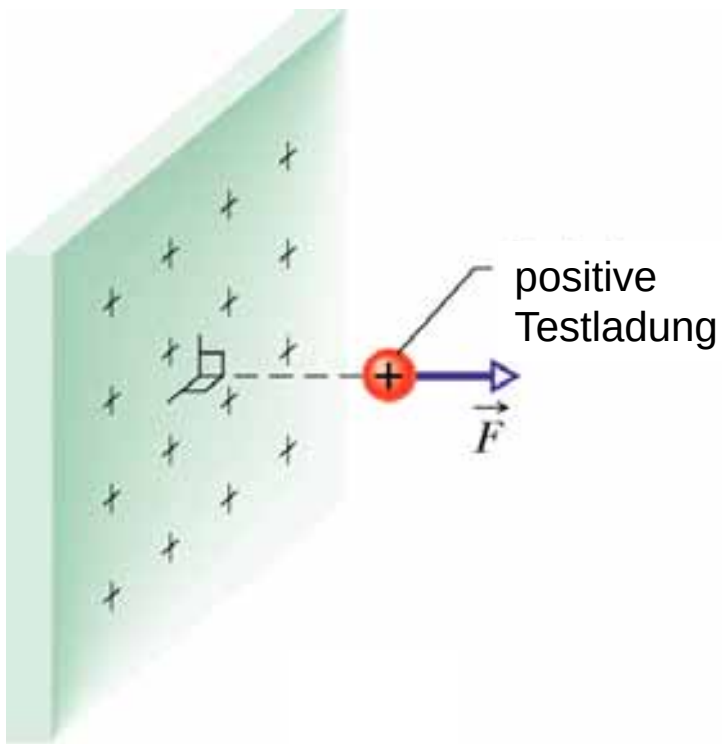
wikipedia.de

zu 3: Elektrische Feldlinien



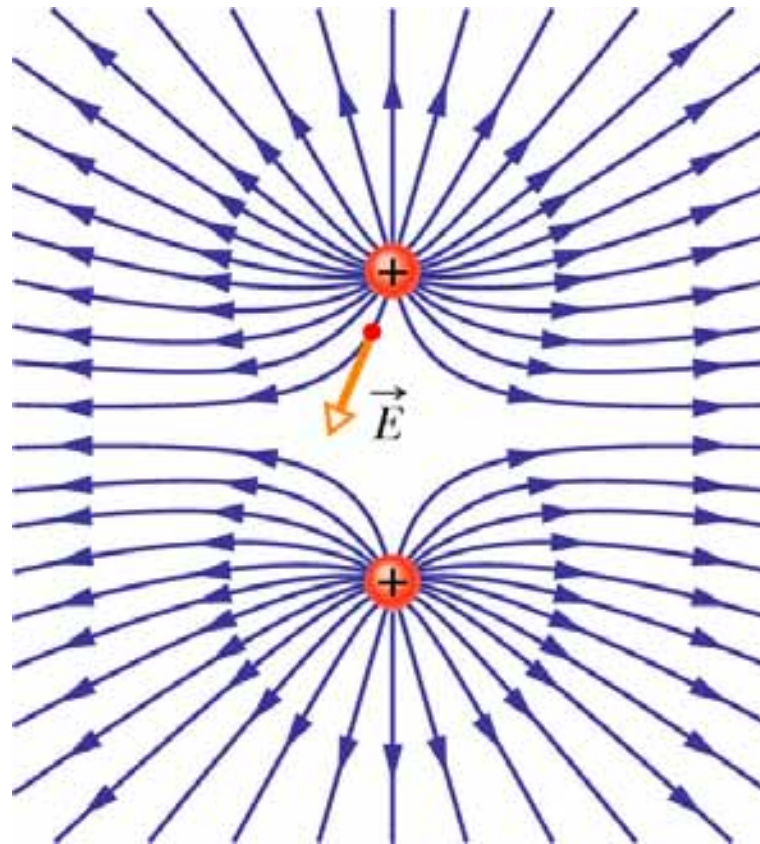
Halliday-Resnick-Walker (2003) Abb. 23.2

zu 3: Elektrische Feldlinien



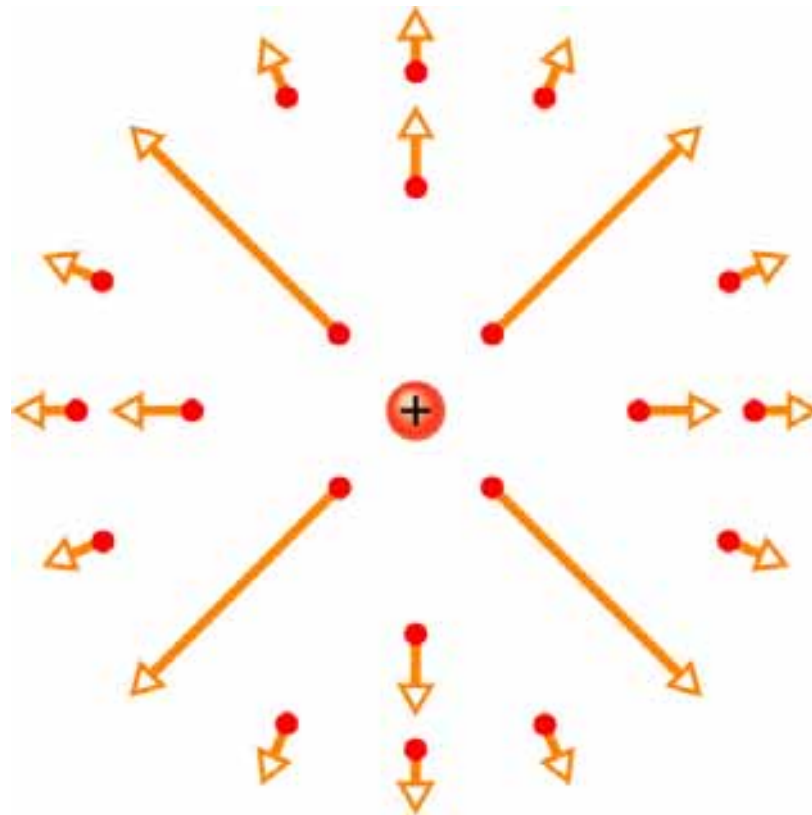
Halliday-Resnick-Walker (2003) Abb. 23.3

zu 3: Elektrische Feldlinien



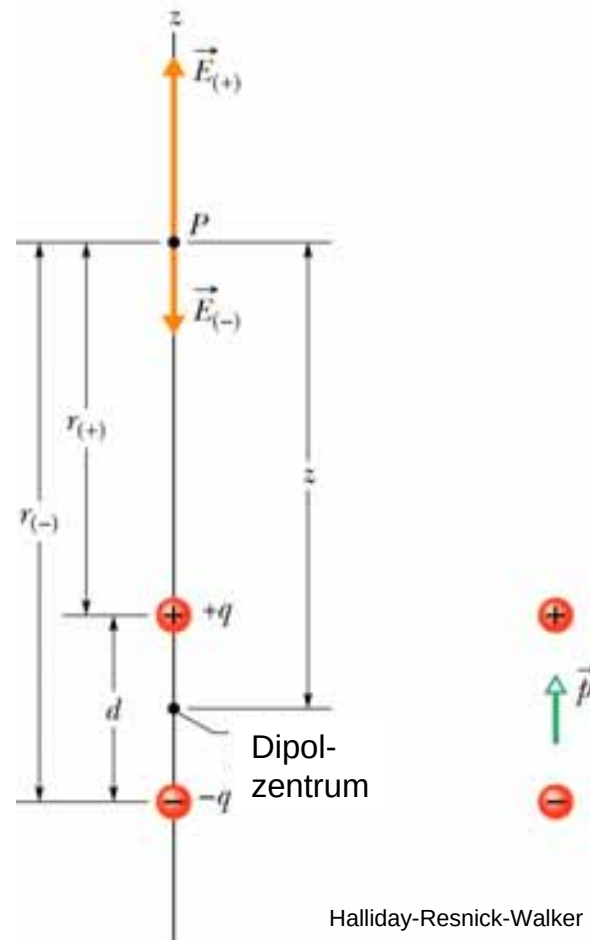
Halliday-Resnick-Walker (2003) Abb. 23.4

zu 3.1: Elektrisches Feld einer Punktladung



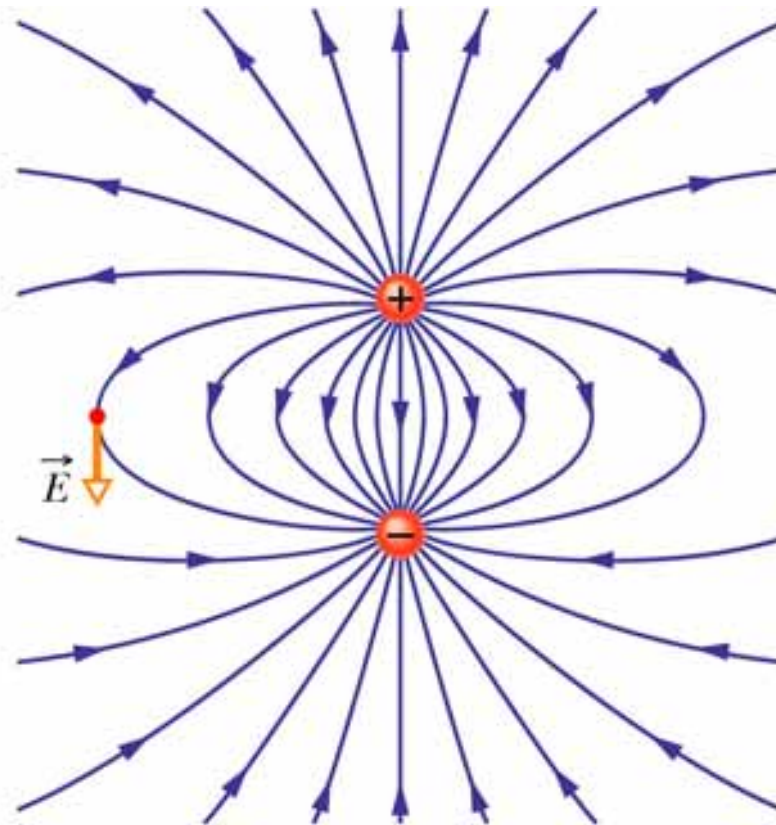
Halliday-Resnick-Walker (2003) Abb. 23.6

zu 3.1: Elektrisches Feld eines Dipols



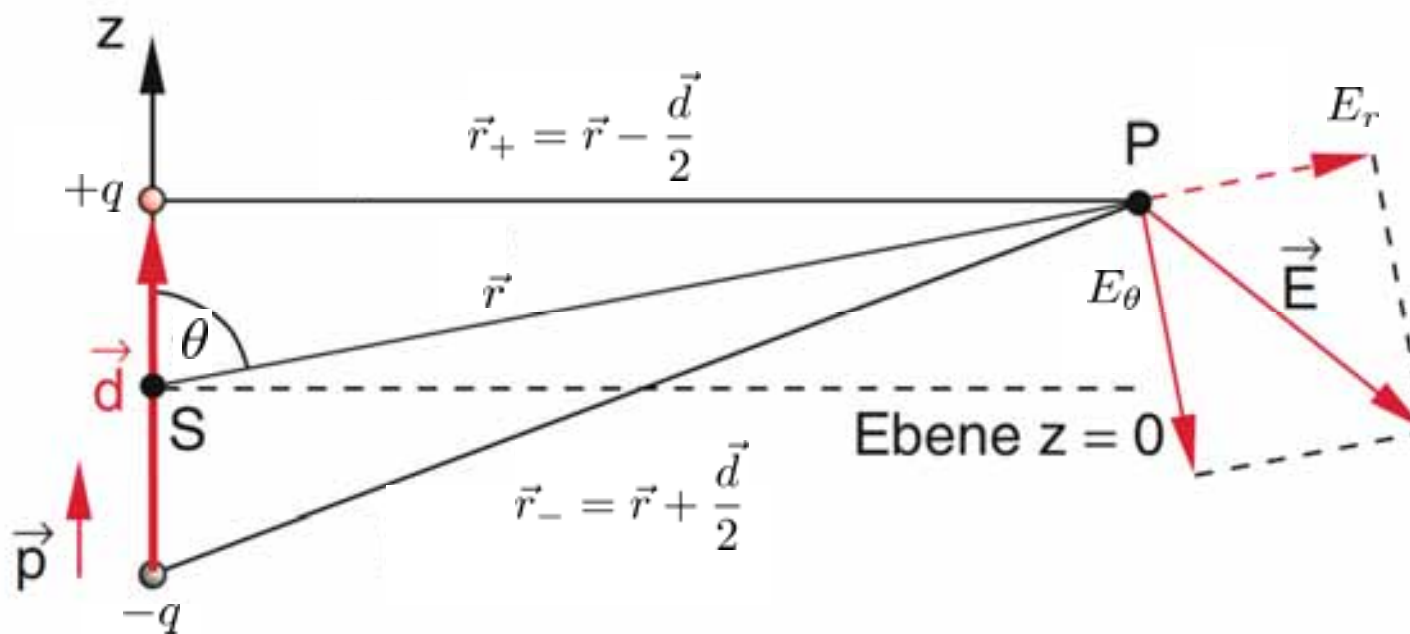
Halliday-Resnick-Walker (2003) Abb. 23.5

zu 3.1: Elektrisches Feld eines Dipols

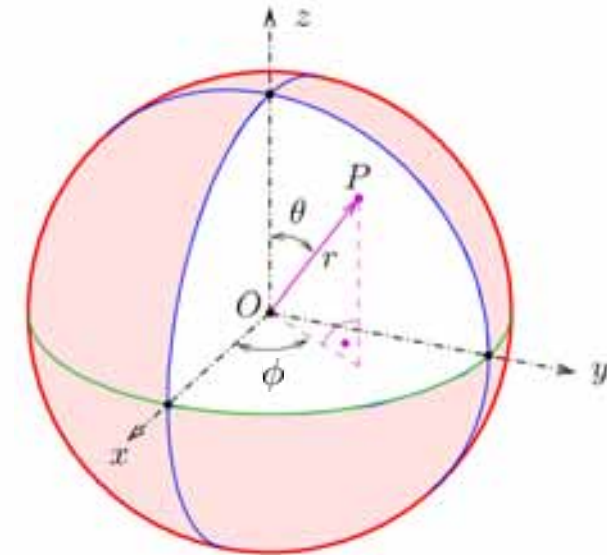


Halliday-Resnick-Walker (2003) Abb. 23.5

zu 3.1: Elektrisches Feld eines Dipols

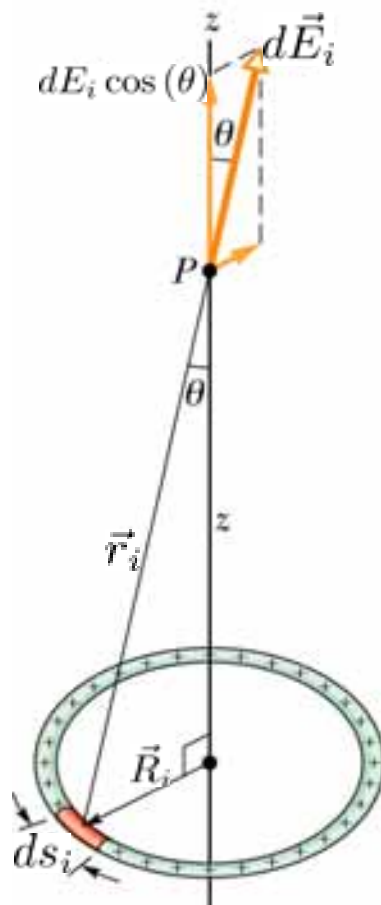


Demtröder – Experimentalphysik 2 (2013), Abb. 1.25.

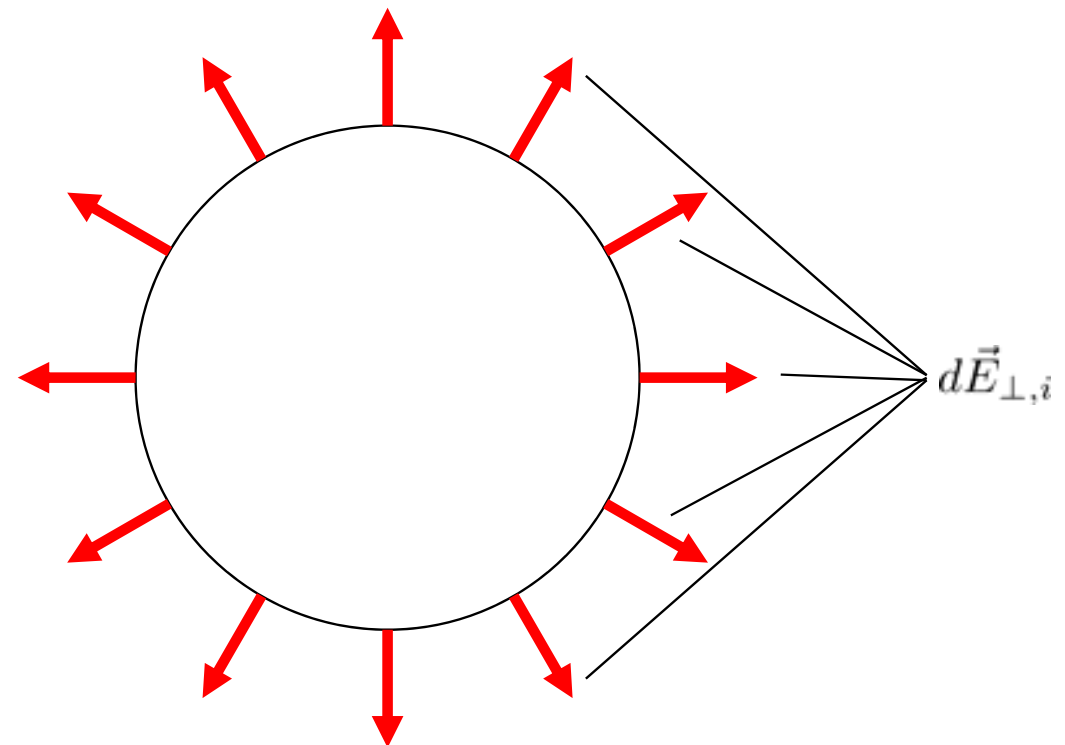


www.wikipedia.org

zu 3.1: Elektrisches Feld einer Linienladung

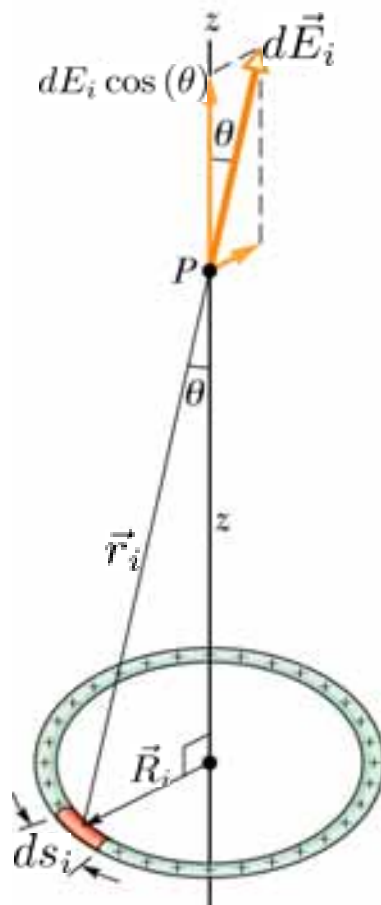


Zur Berechnung von $\vec{E}_\perp$:

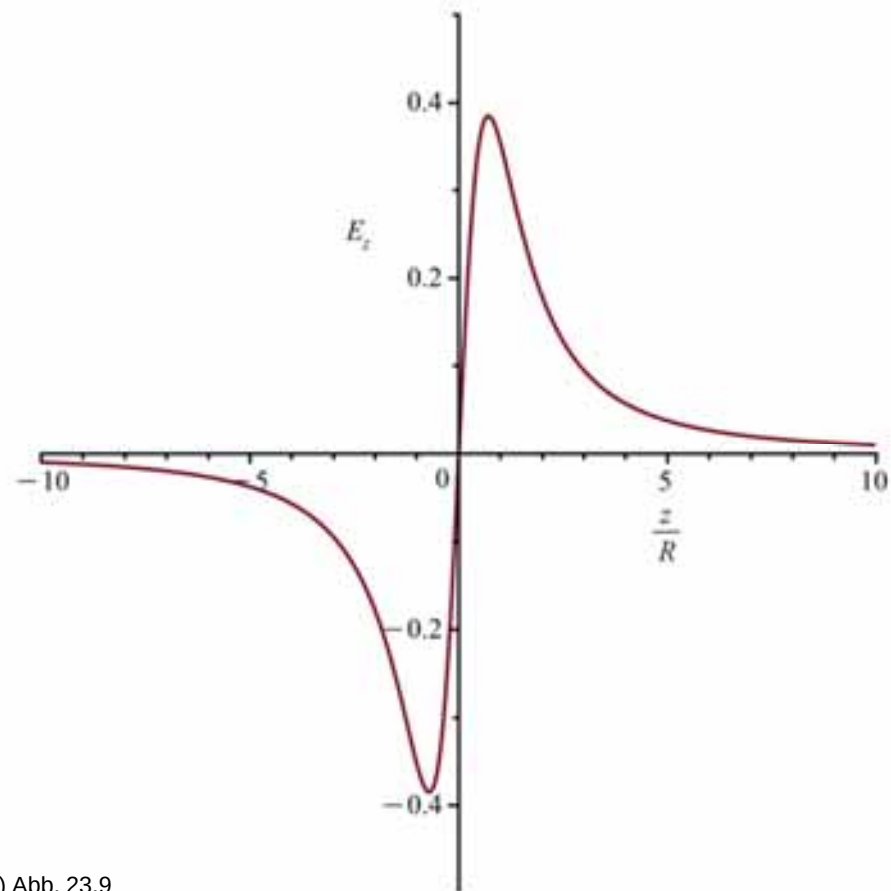


Halliday-Resnick-Walker (2003) Abb. 23.9

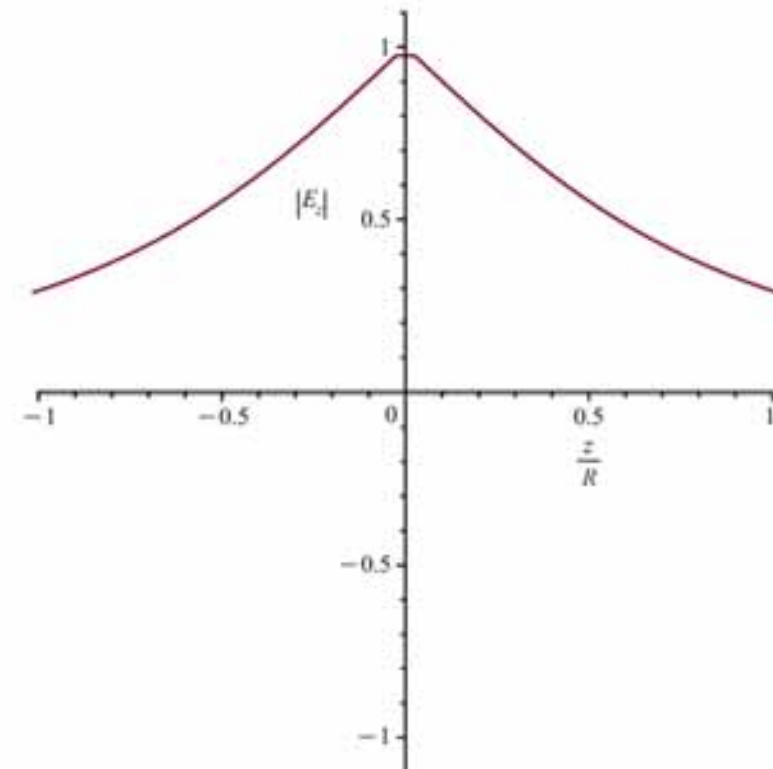
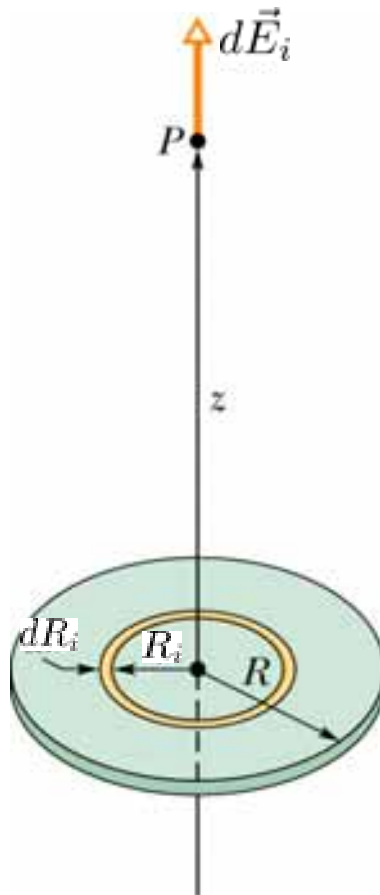
zu 3.1: Elektrisches Feld einer Linienladung



Halliday-Resnick-Walker (2003) Abb. 23.9

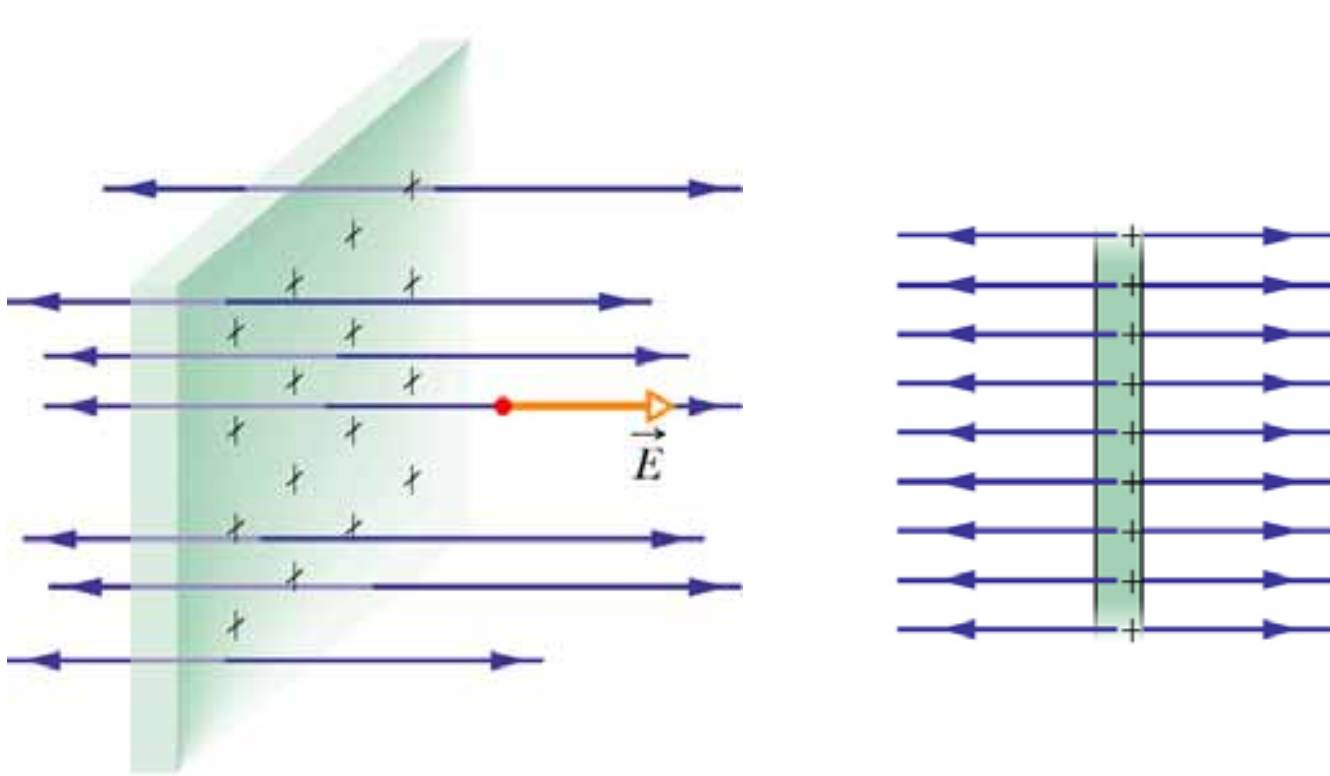


zu 3.1: Elektrisches Feld einer Flächenladung



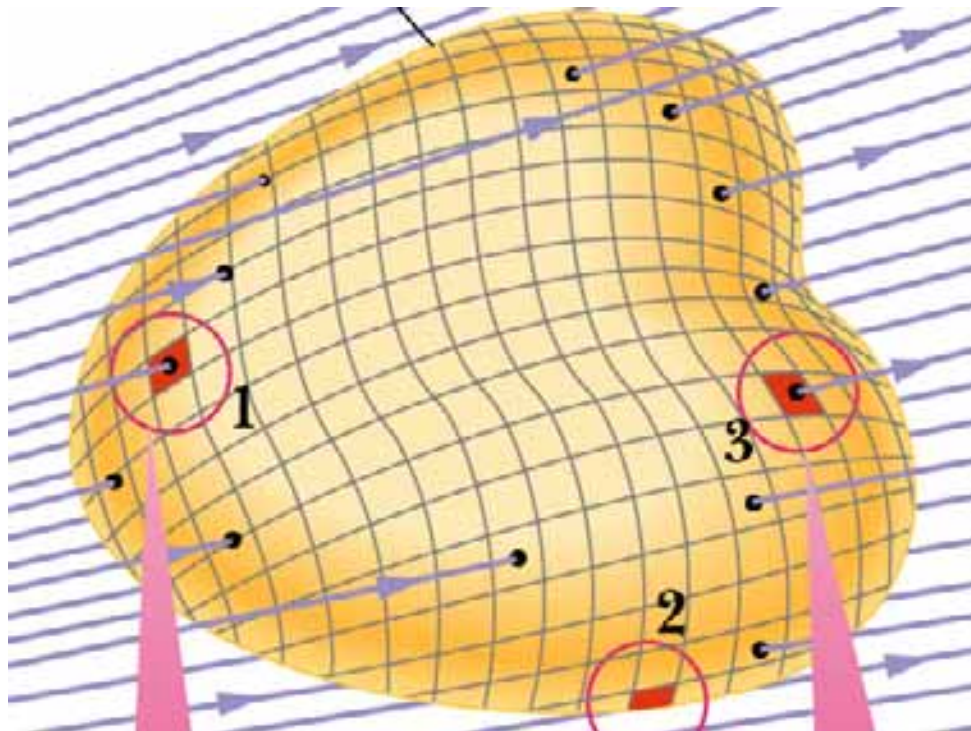
Halliday-Resnick-Walker (2003) Abb. 23.10

zu 3.1: Elektrisches Feld einer Flächenladung



Halliday-Resnick-Walker (2003) Abb. 23.3

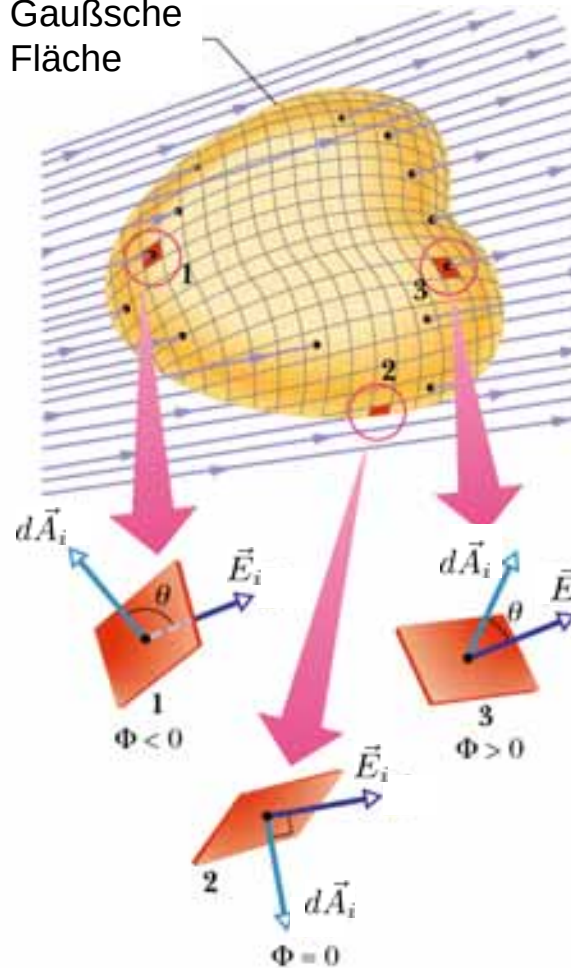
zu 3.2: Gaußsche Fläche



Halliday-Resnick-Walker (2003) Abb. 24.3

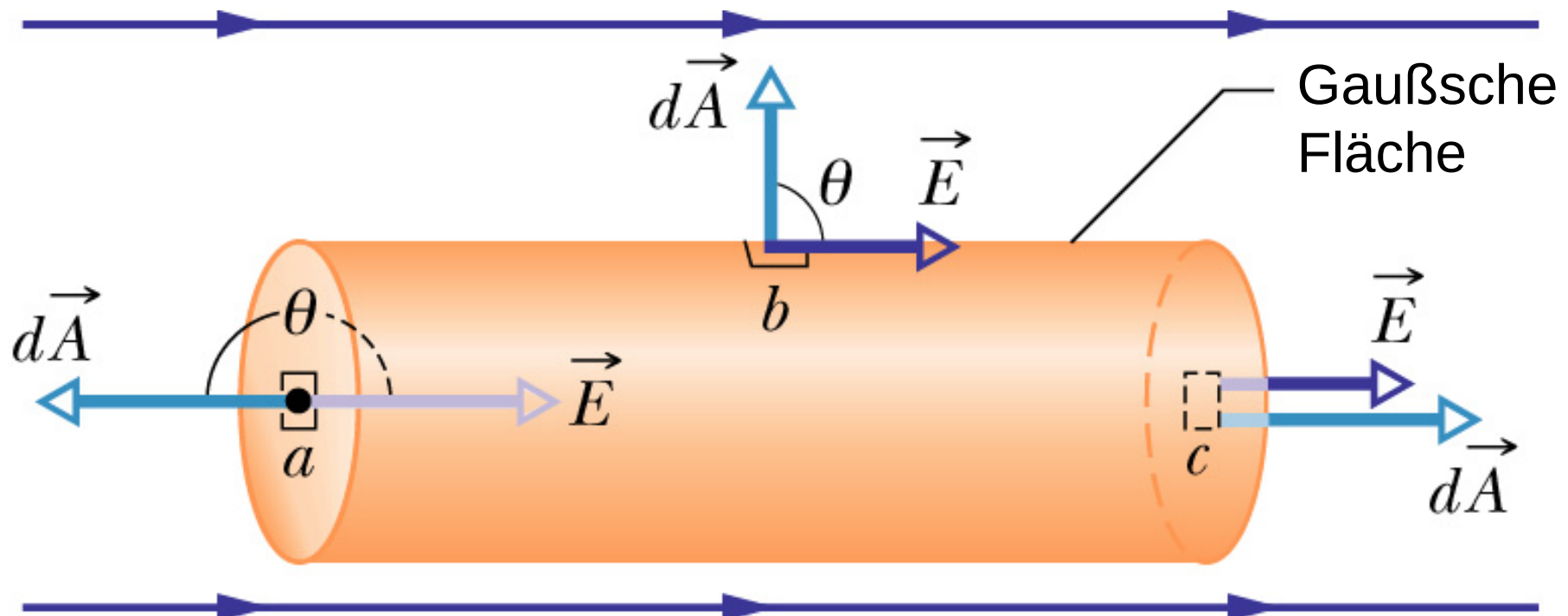
zu 3.2: Gaußsche Fläche und elektrischer Fluss

Gaußsche
Fläche



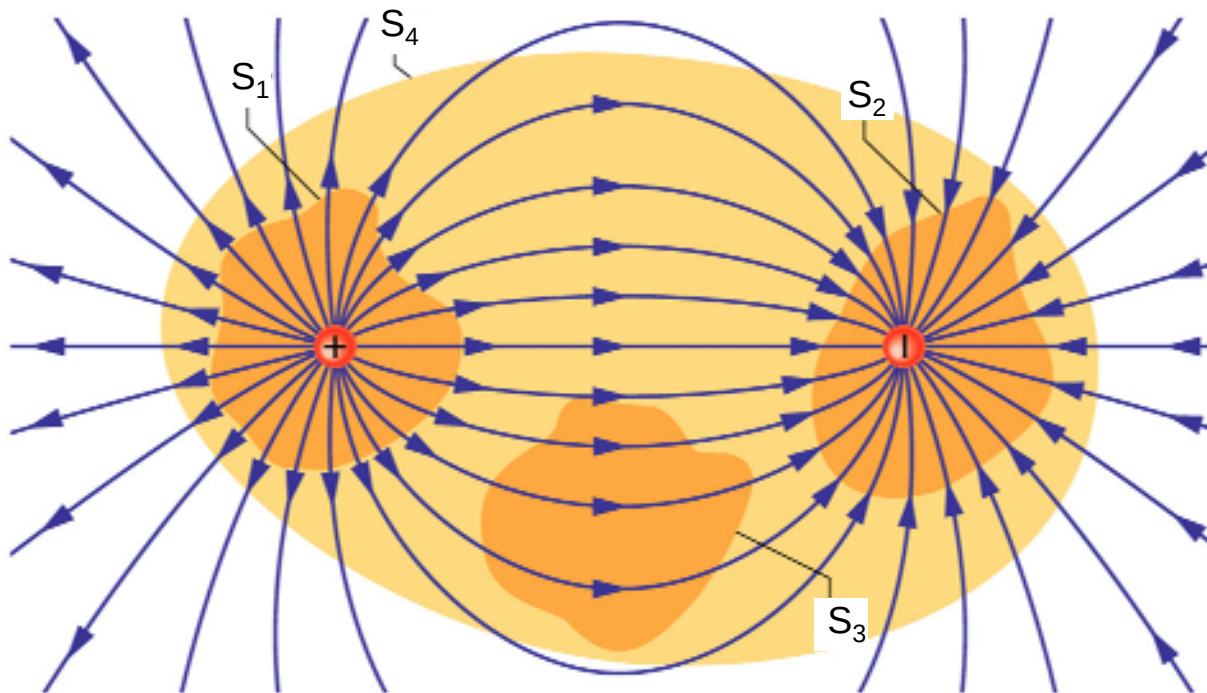
Halliday-Resnick-Walker (2003) Abb. 24.3

zu 3.2: Gaußsche Fläche und elektrischer Fluss



Halliday-Resnick-Walker (2003) Abb. 24.4

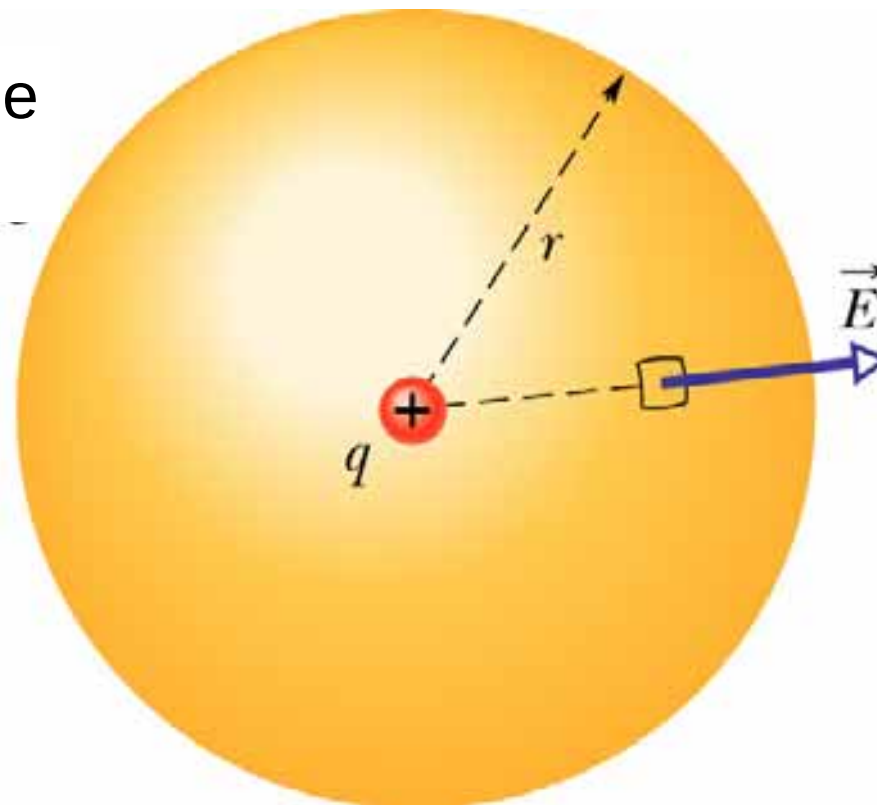
zu 3.2: Der Gaußsche Satz



Halliday-Resnick-Walker (2003) Abb. 24.6

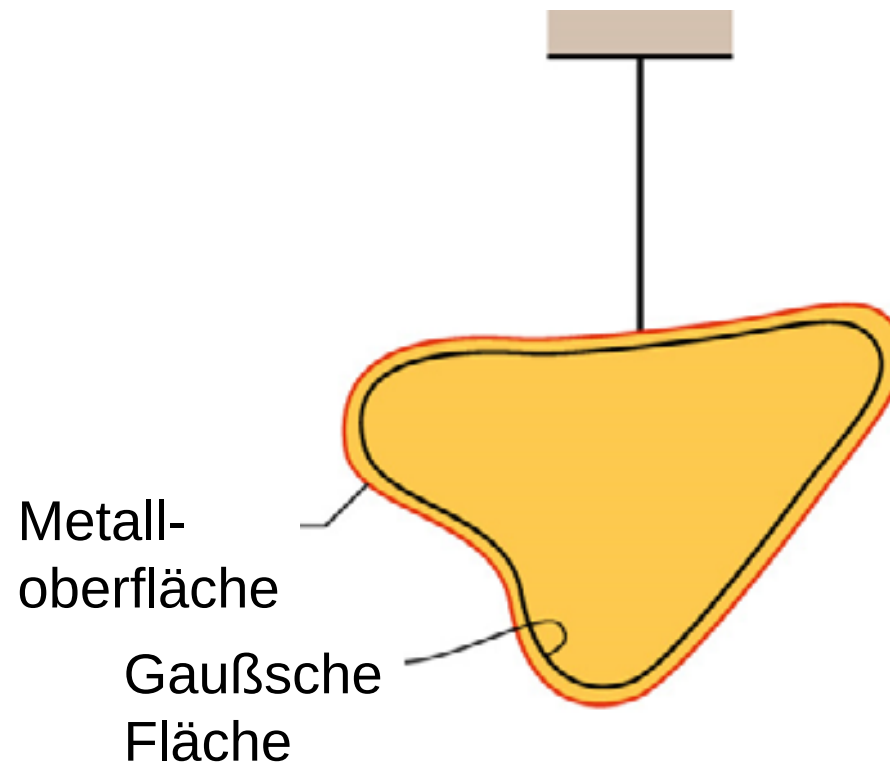
zu 3.2: Der Gaußsche Satz

Gaußsche
Fläche



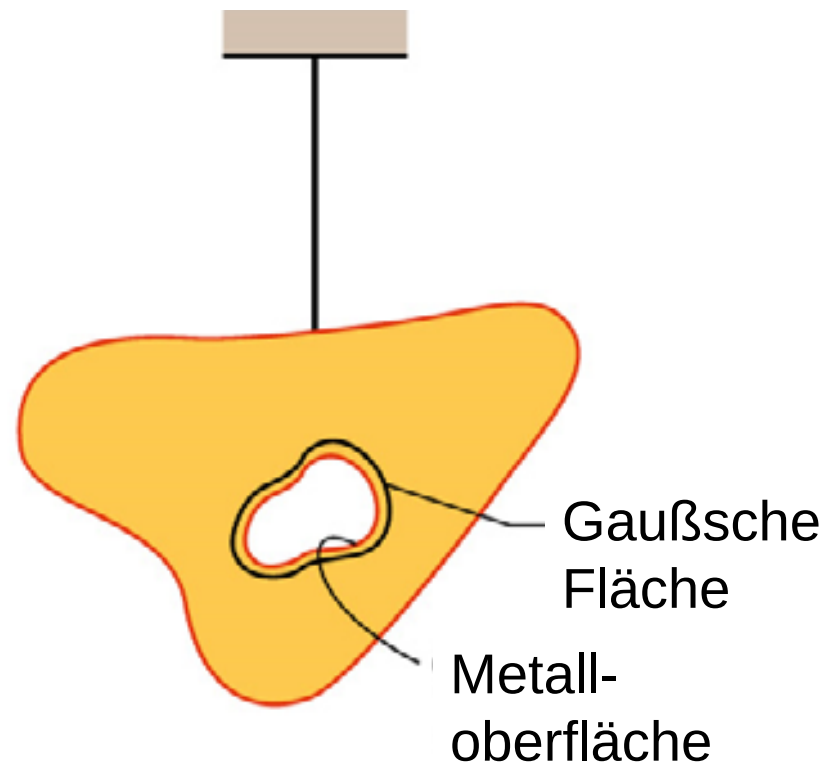
Halliday-Resnick-Walker (2003) Abb. 24.8

zu 3.2: Der Gaußsche Satz



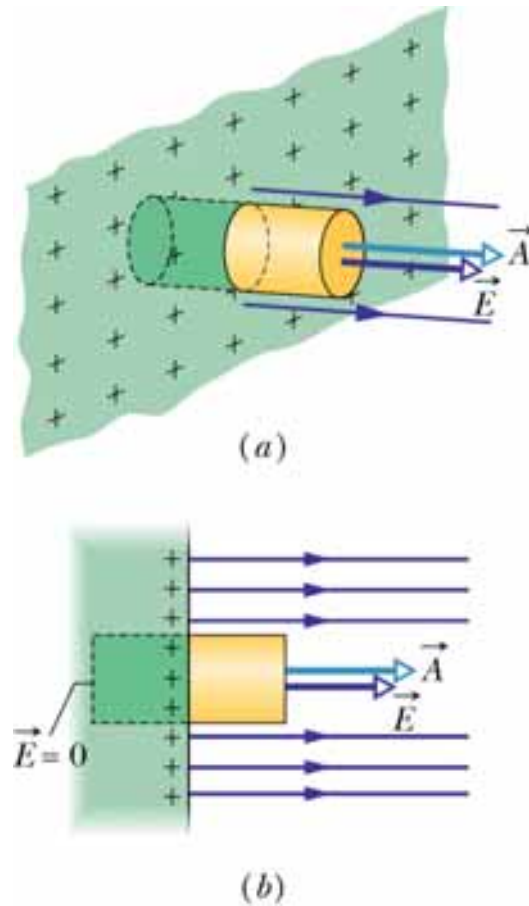
Halliday-Resnick-Walker (2003) Abb. 24.9

zu 3.2: Der Gaußsche Satz



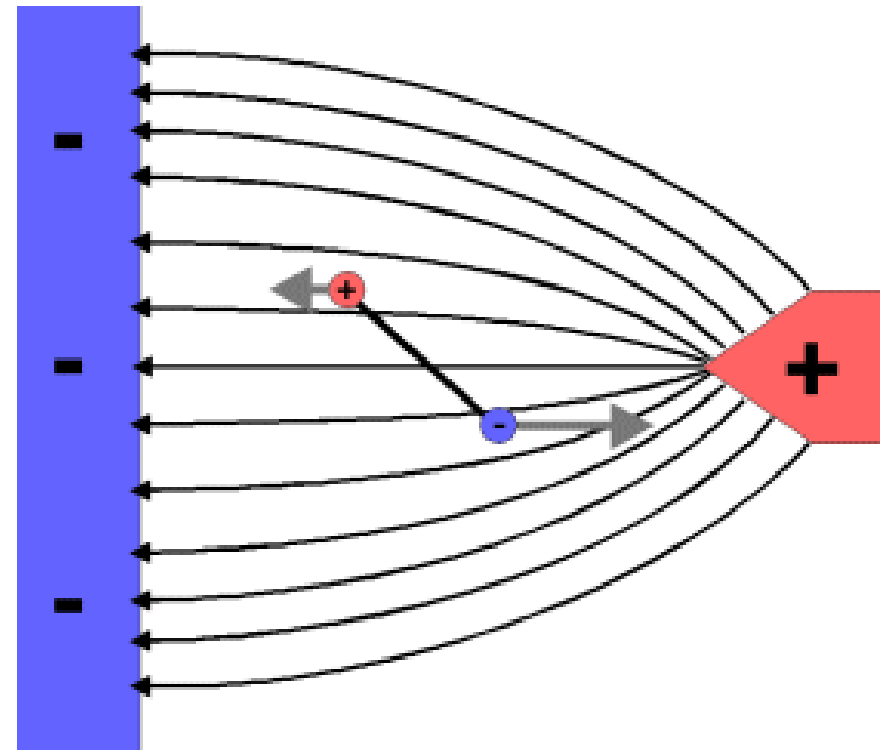
Halliday-Resnick-Walker (2003) Abb. 24.9

zu 3.2: Das äußere elektrische Feld



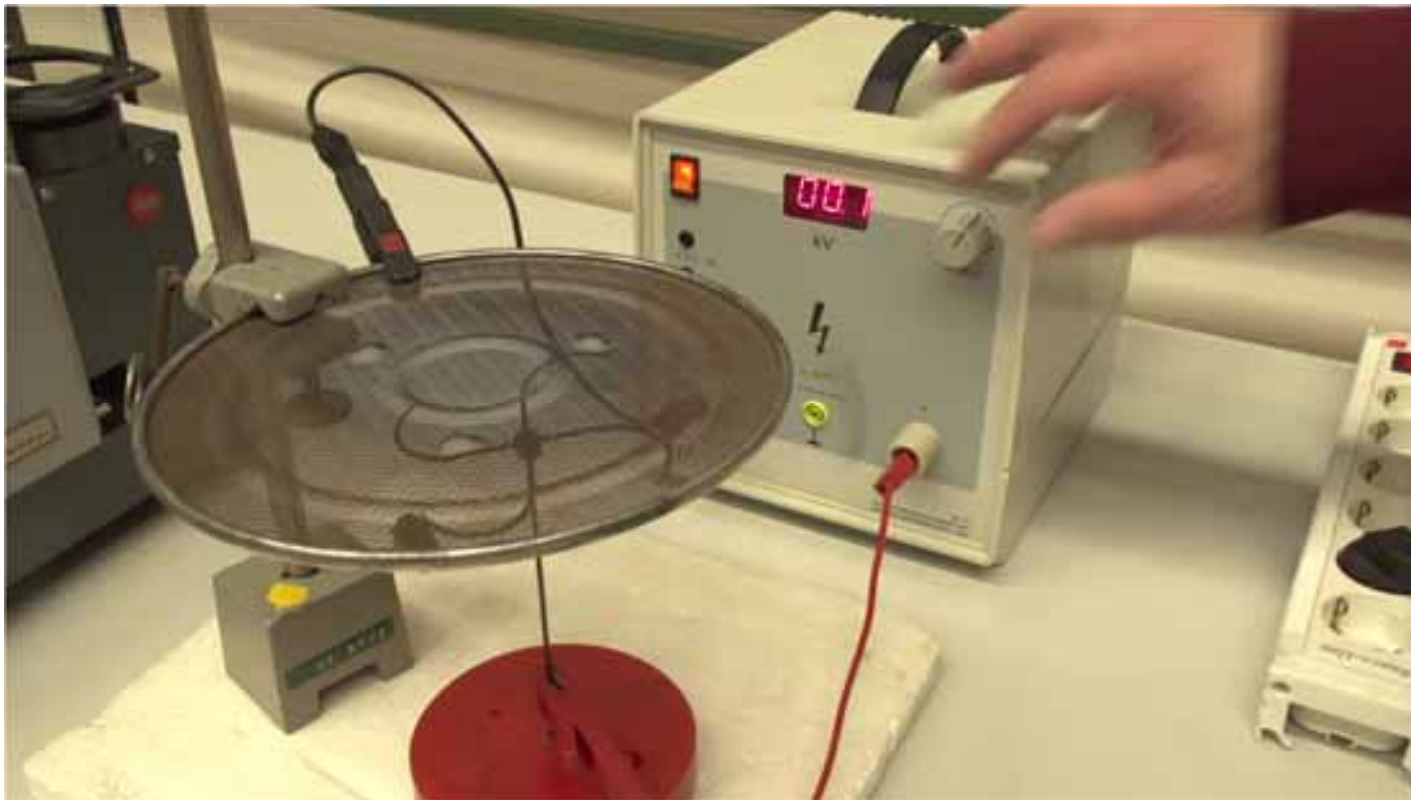
Halliday-Resnick-Walker (2003) Abb. 24.10

zu 3.2: Spitzenentladung

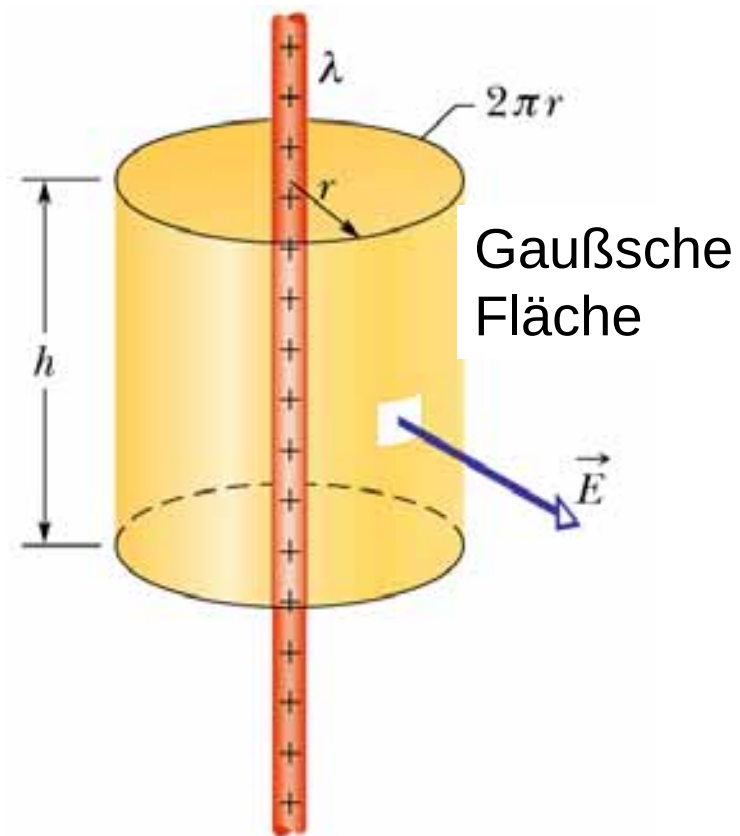


leifiphysik.de

zu 3.2: Segnerrad im elektrischen Feld

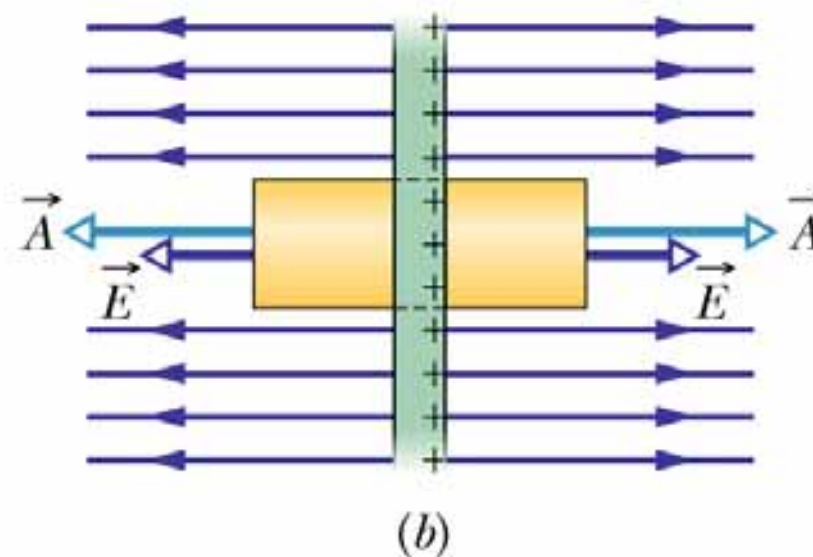
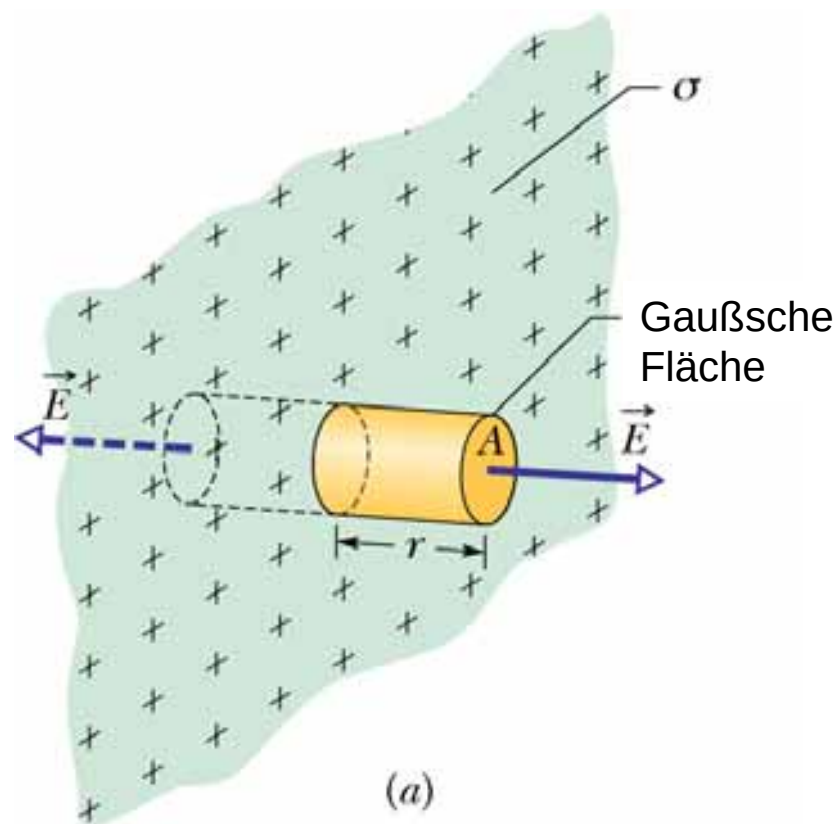


zu 3.2: Lineare Ladungsverteilung



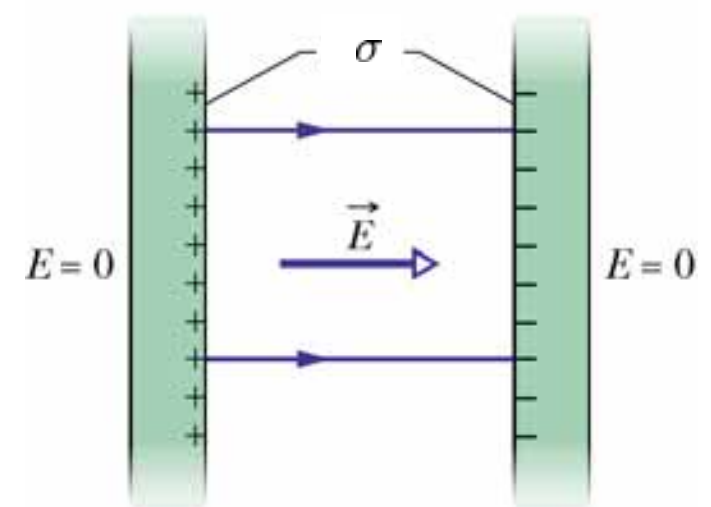
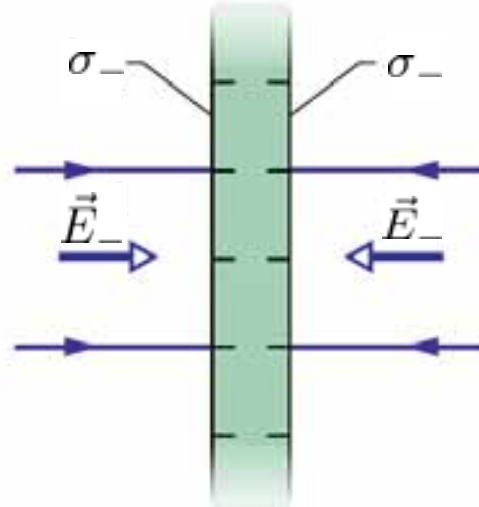
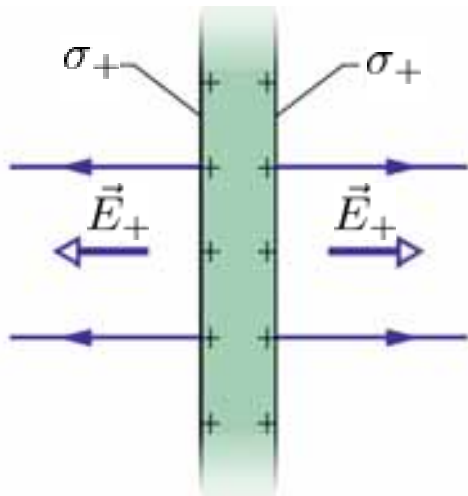
Halliday-Resnick-Walker (2003) Abb. 24.12

zu 3.2: Nichtleitende Folie



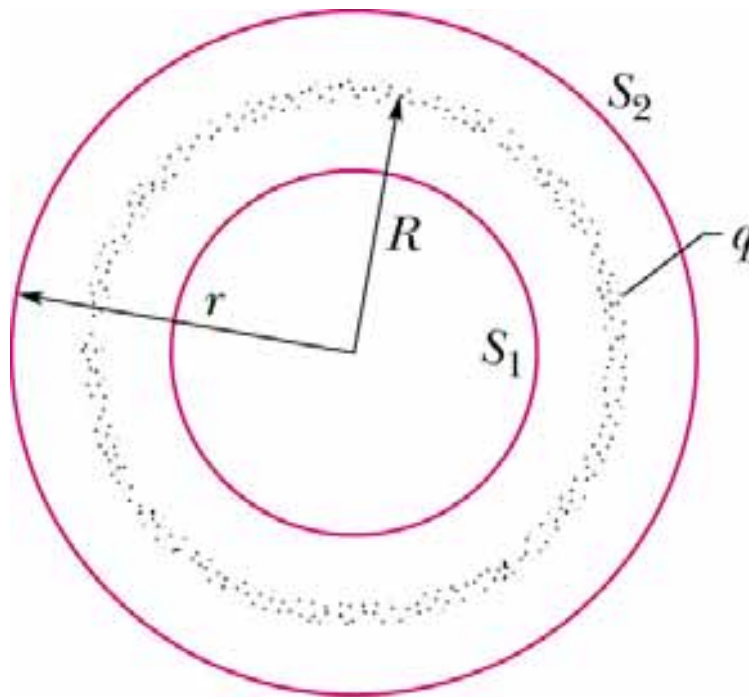
Halliday-Resnick-Walker (2003) Abb. 24.15

zu 3.2: Plattenkondensator



Halliday-Resnick-Walker (2003) Abb. 24.16

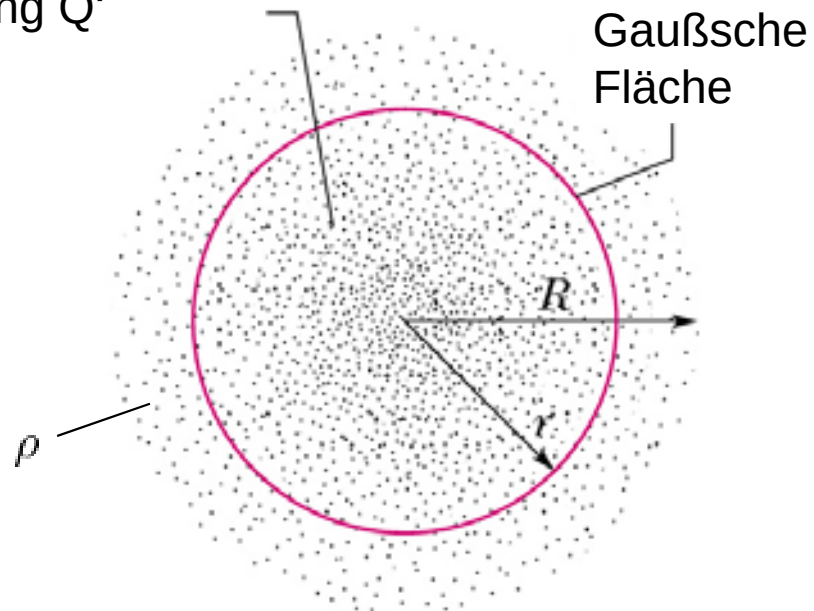
zu 3.2: Geladene Kugelschale



Halliday-Resnick-Walker (2003) Abb. 24.18

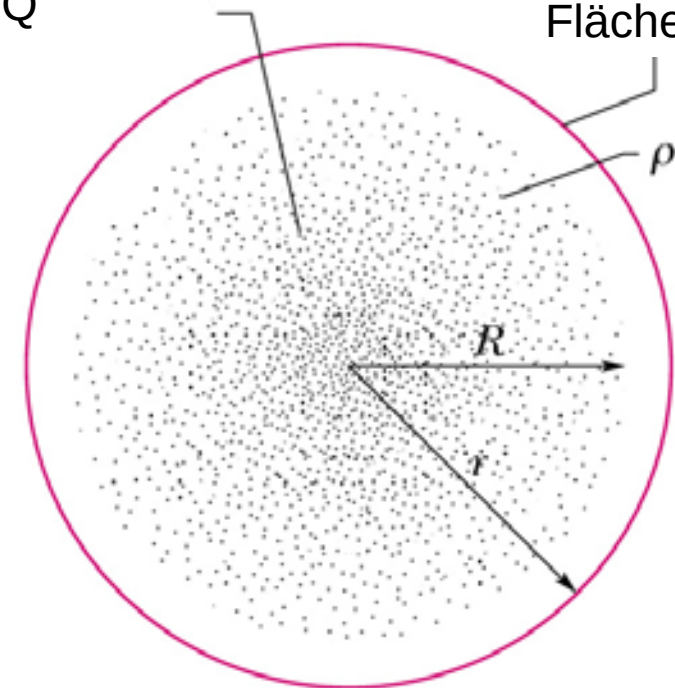
zu 3.2: Geladene Vollkugel

Eingeschlossene
Ladung Q'



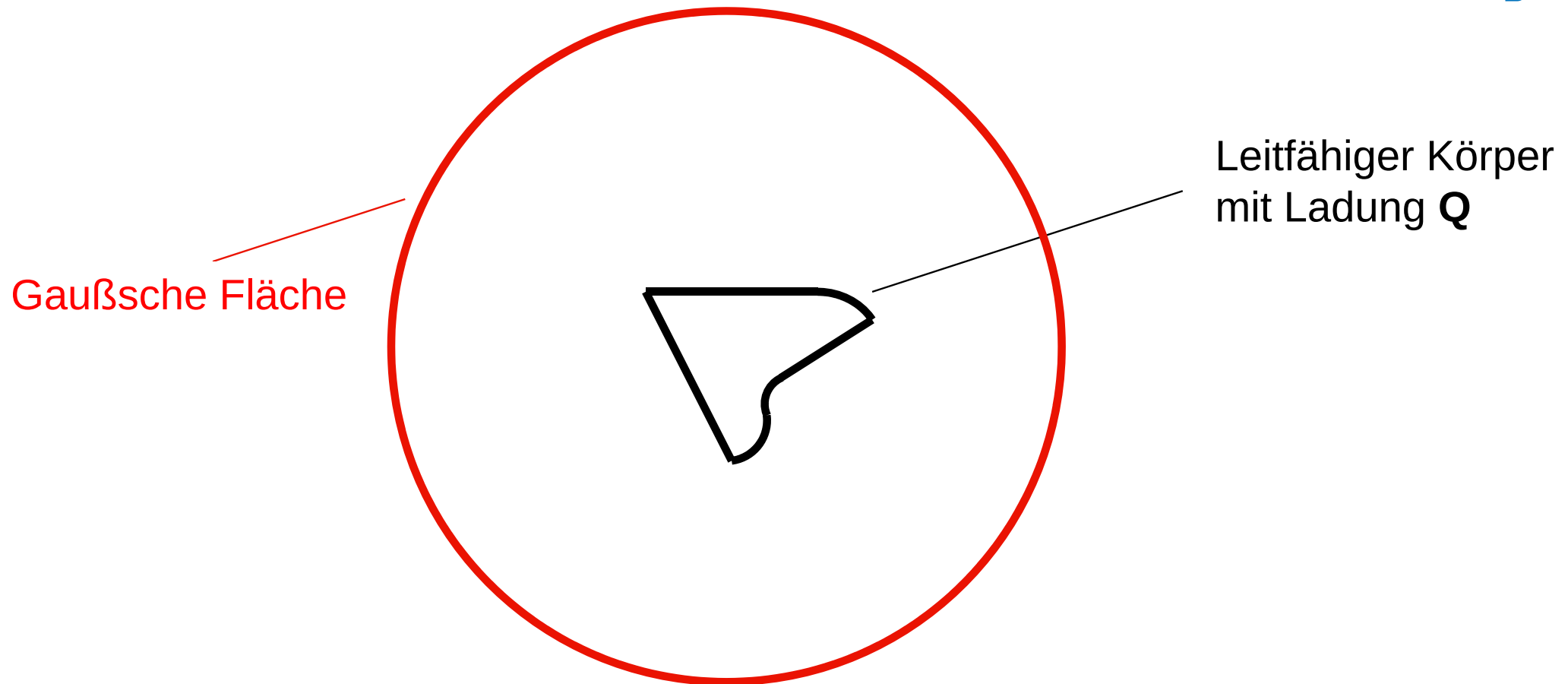
Eingeschlossene
Ladung Q

Gaußsche
Fläche

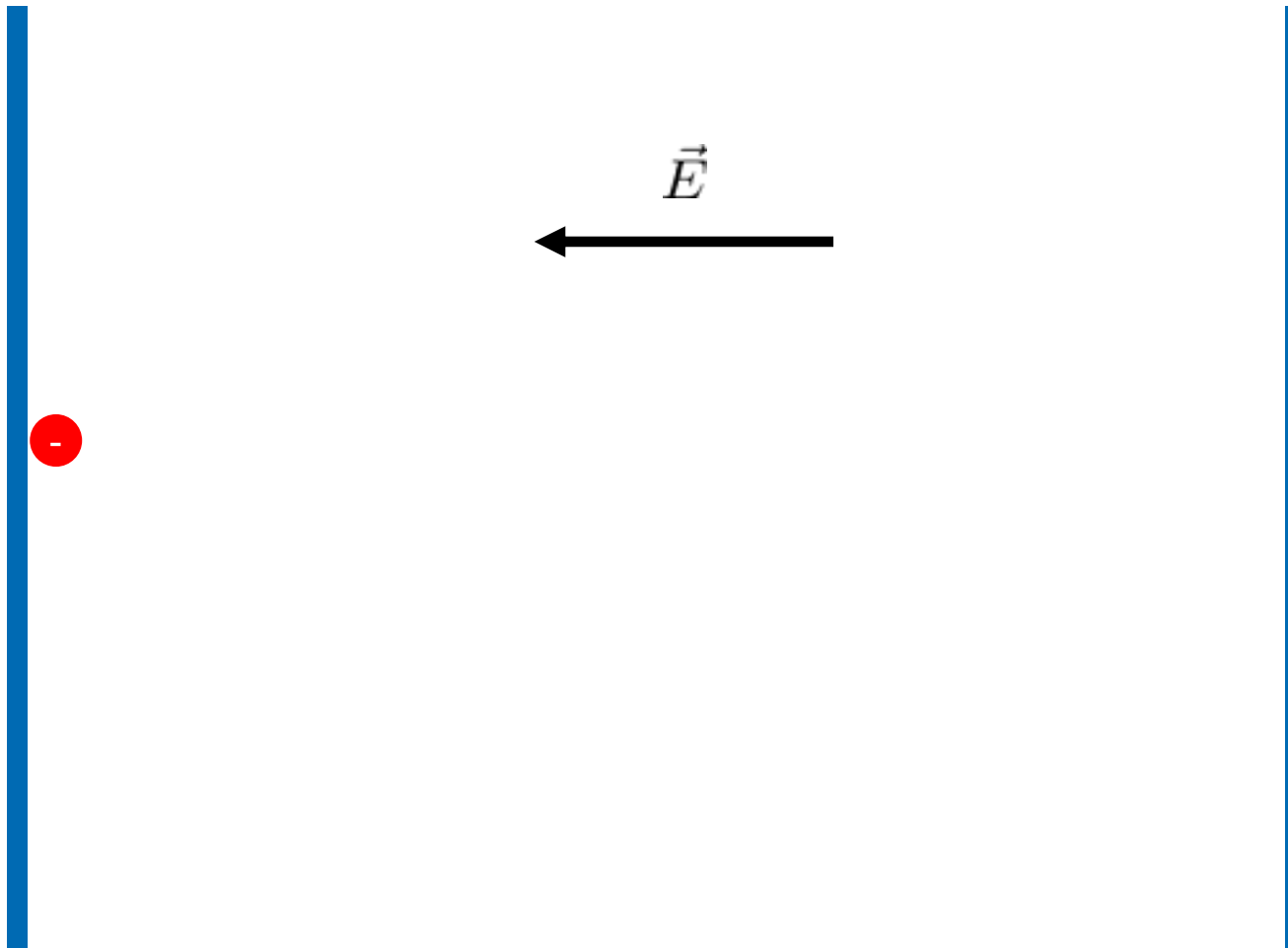


Halliday-Resnick-Walker (2003) Abb. 24.19

zu 3.2: Allgemeine Ladungsverteilungen



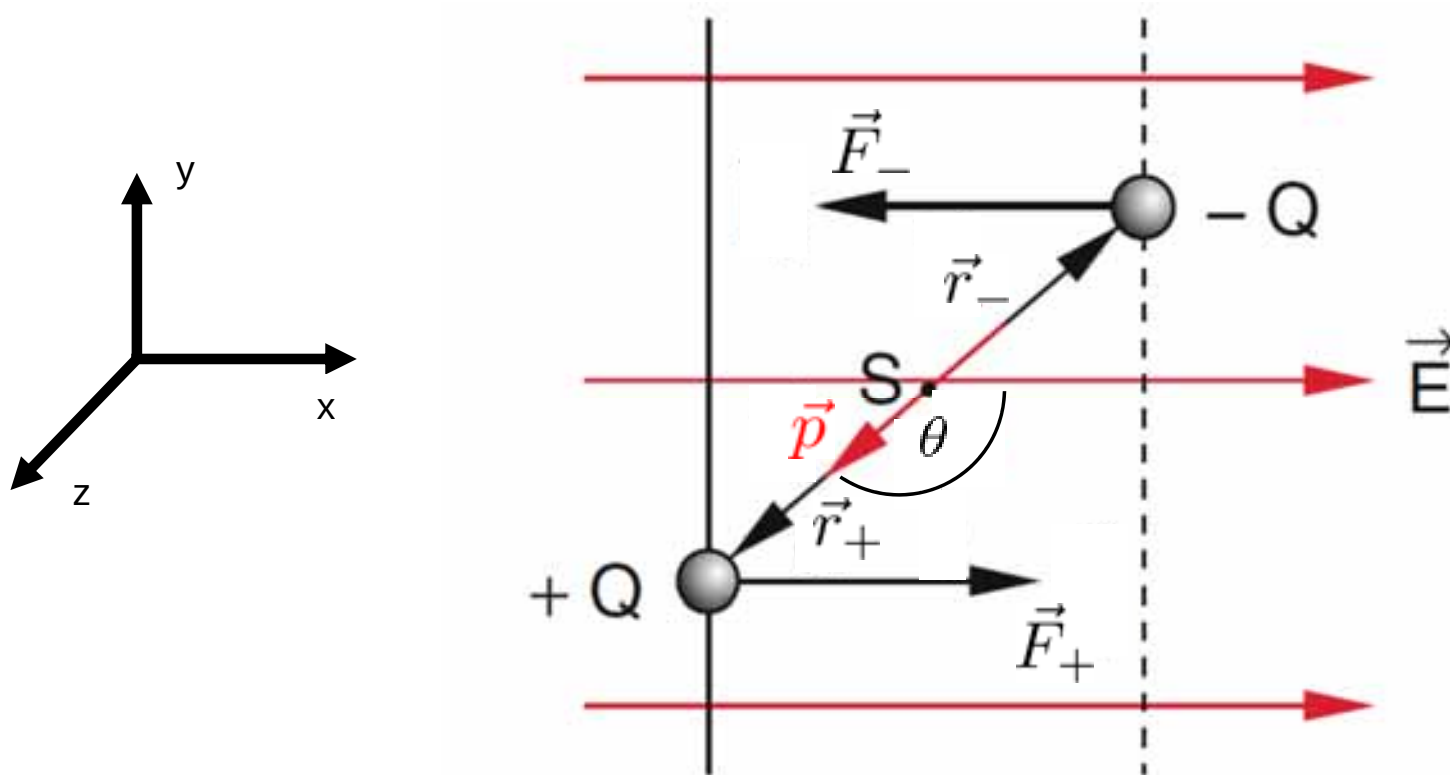
zu 3.3: Punktladung im elektrischen Feld



zu 3.3: Punktladung im elektrischen Feld

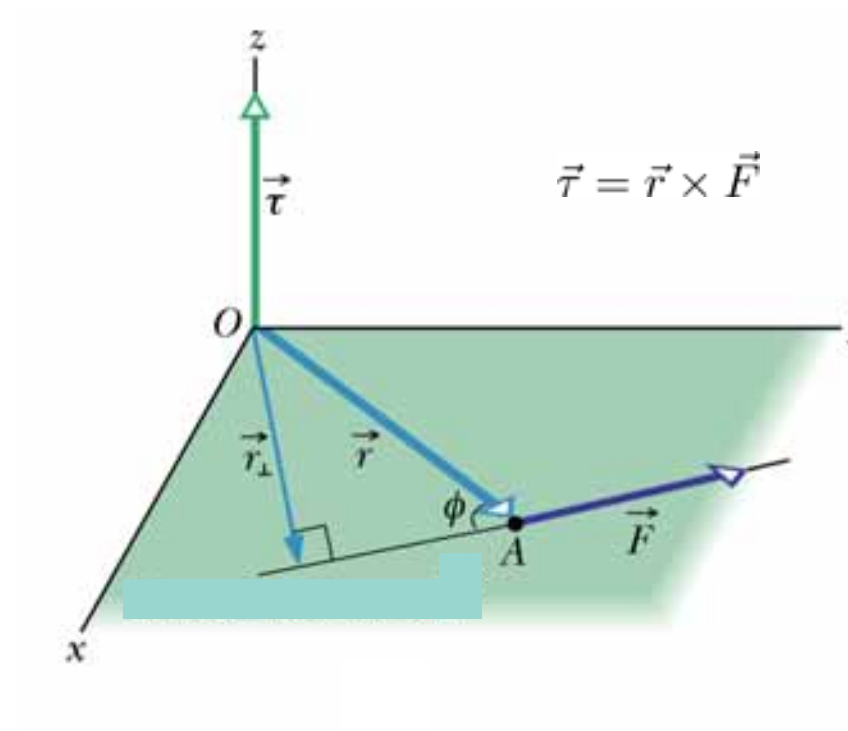


zu 3.3: Dipole im elektrischen Feld



Demtröder – Experimentalphysik 2 (2013), Abb. 1.26.

zu 3.3: Drehmoment

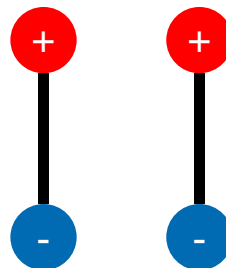


Halliday-Resnick-Walker (2003) Abb. 12.9

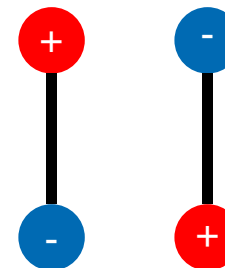
zu 3.3: Wechselwirkung zwischen Dipolen



Anziehende Kraft

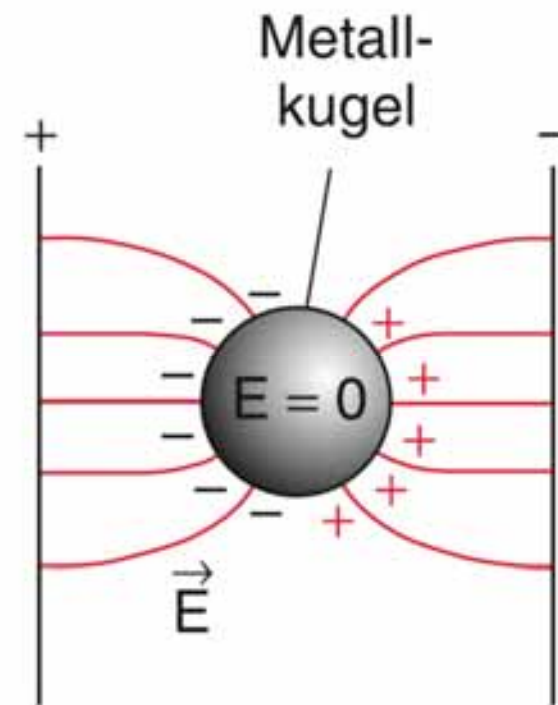
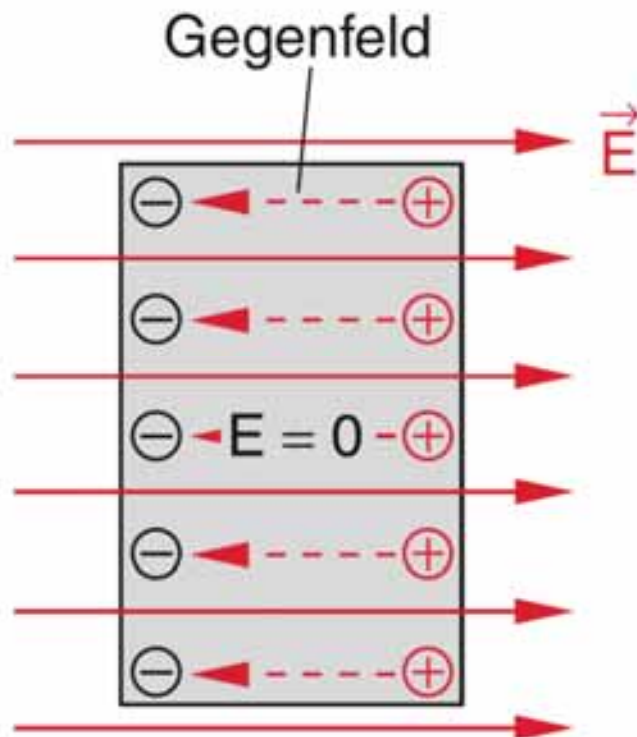


Abstoßende Kraft



Anziehende Kraft

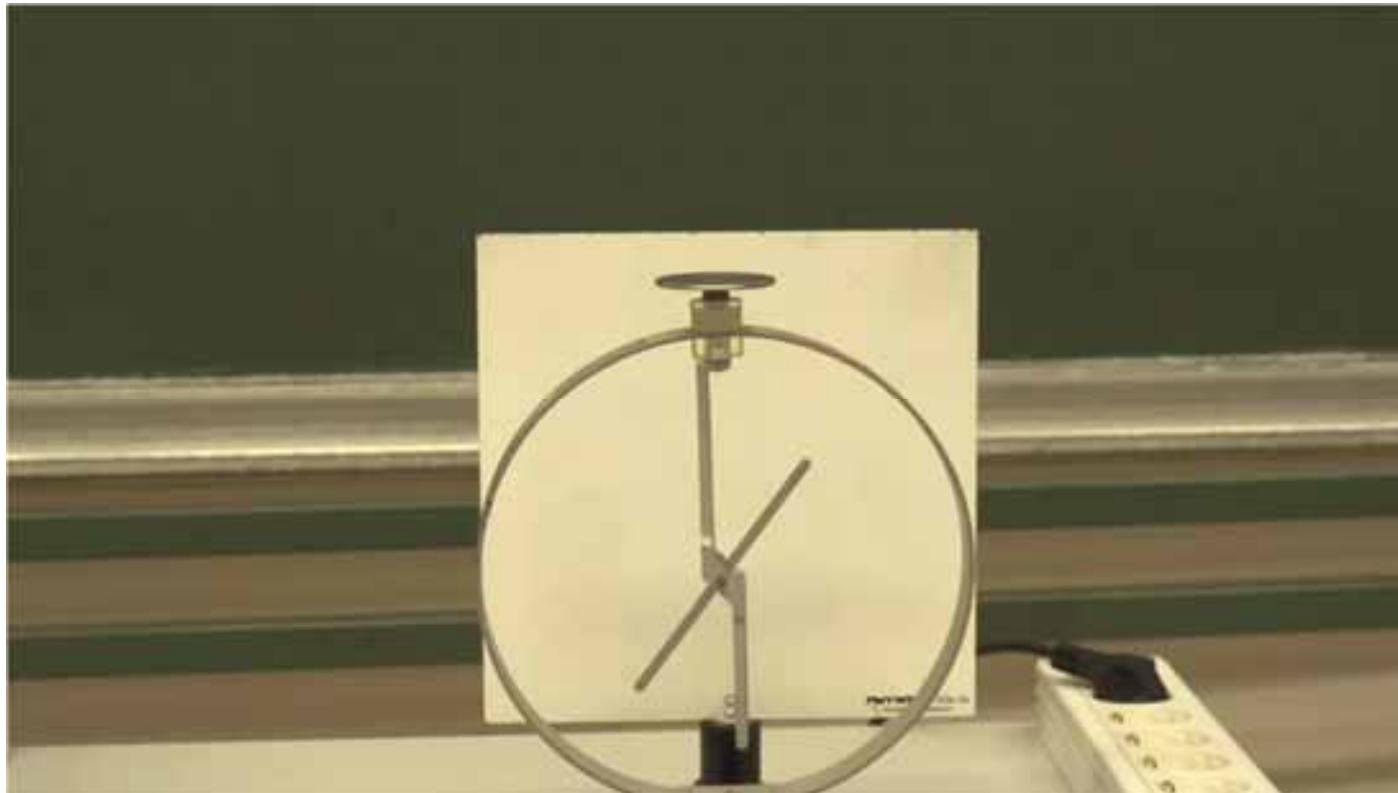
zu 3.3: Influenz



Demtröder – Experimentalphysik 2 (2013), Abb. 1.29.

zu 3.3: Influenz

Influenz am Elektroskop

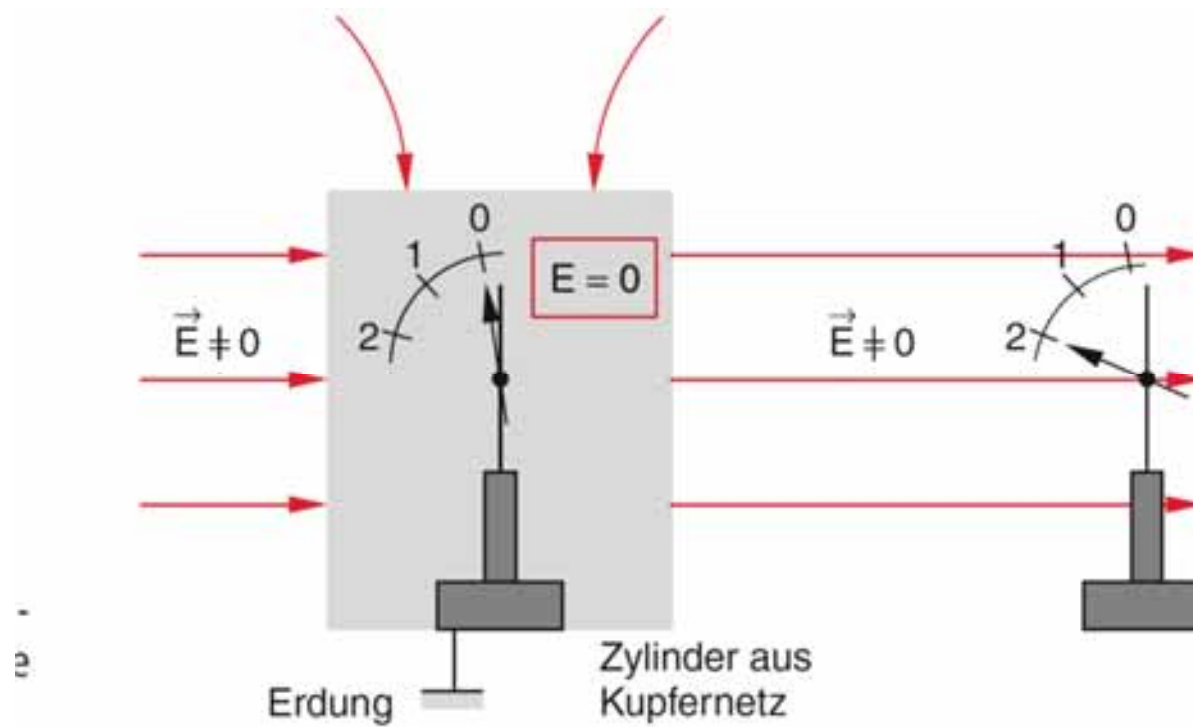


zu 3.3: Influenz

Faraday-Käfig



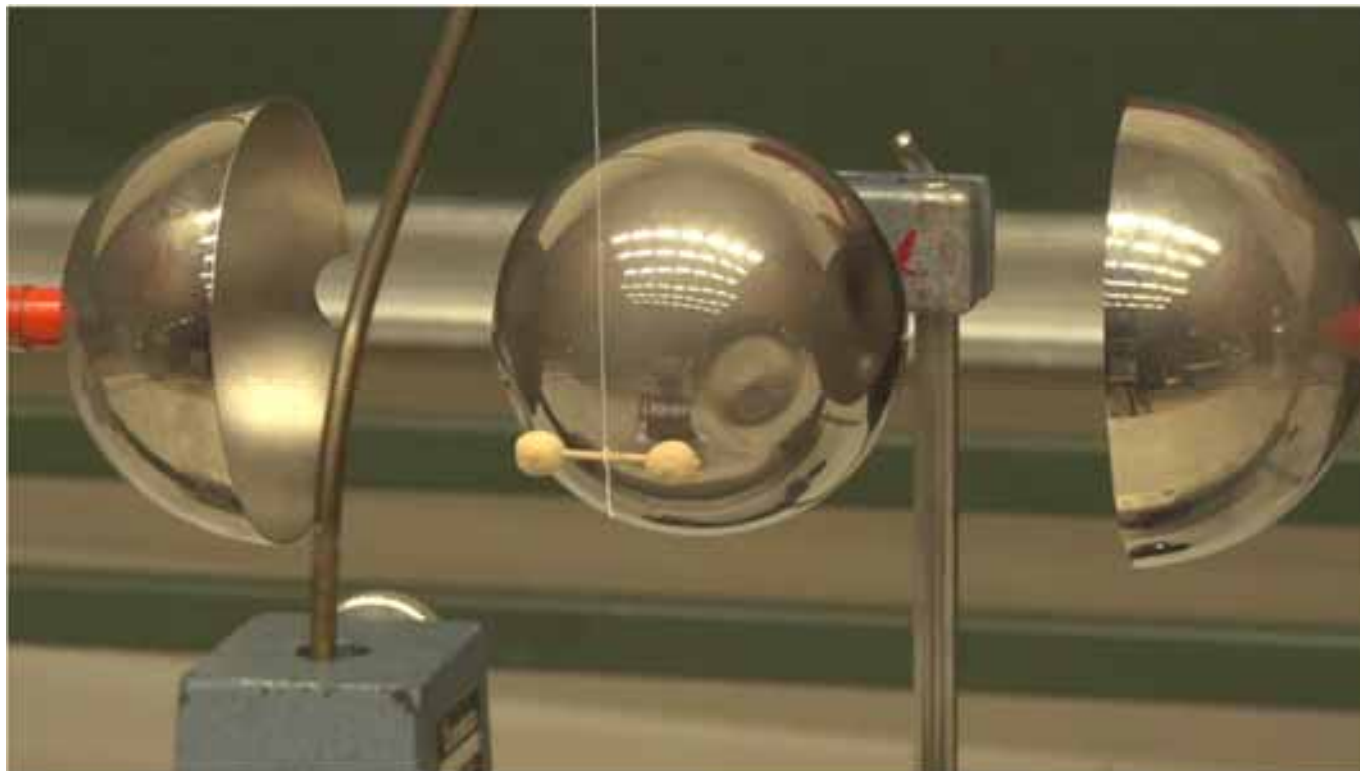
zu 3.3: Influenz



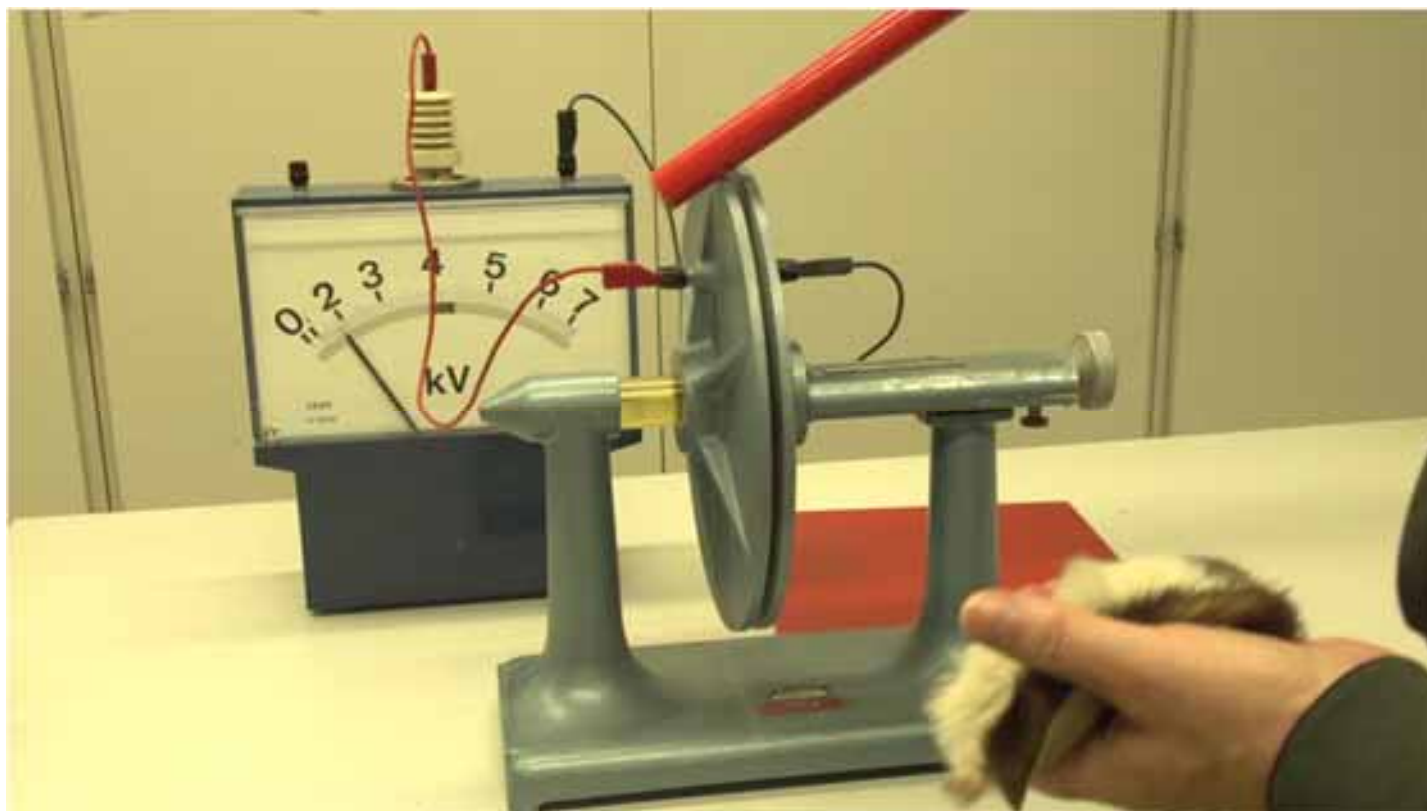
Demtröder – Experimentalphysik 2 (2013), Abb. 1.31.

zu 3.3: Influenz

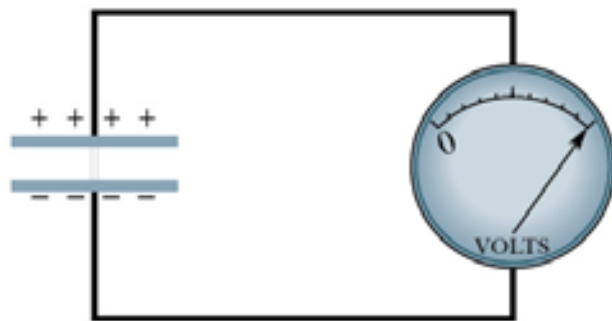
Cavendish-Halbkugeln



zu 3.3: Dielektrikum im Kondensator



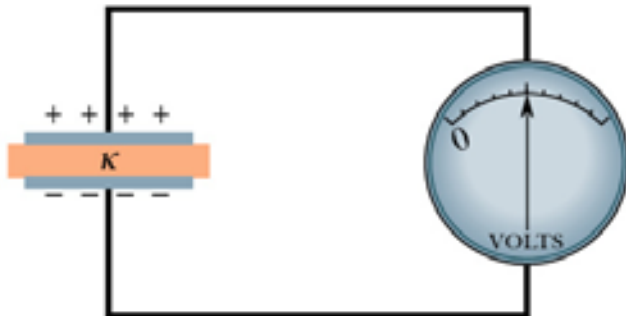
zu 3.3: Dielektrikum im Kondensator



Q konstant

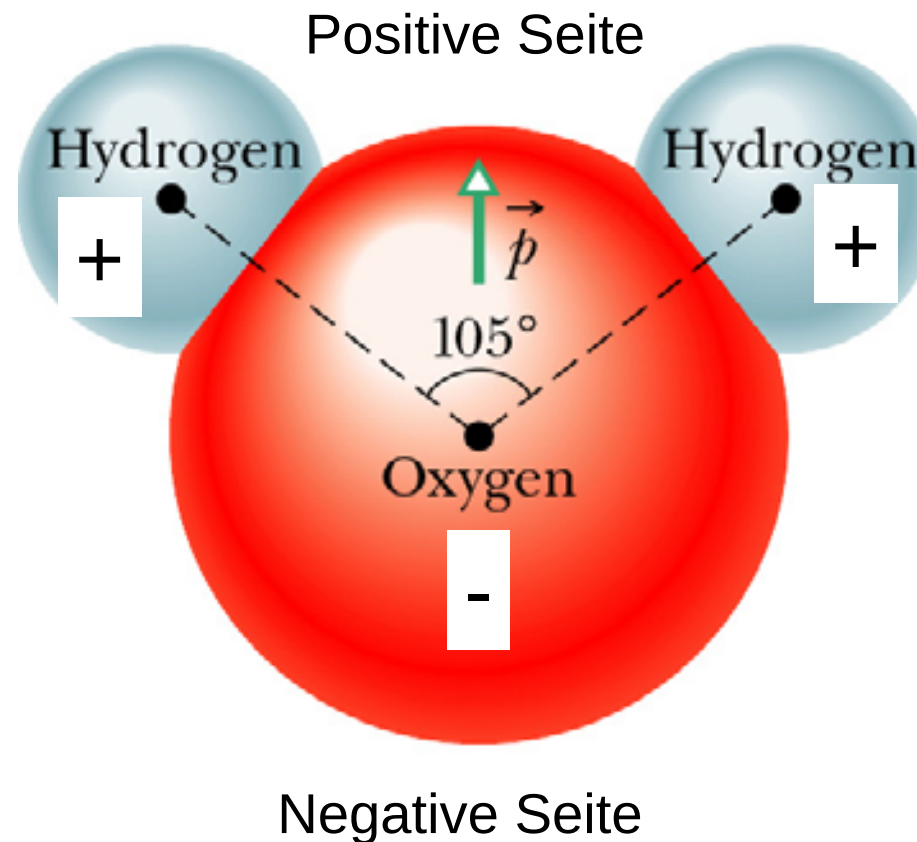


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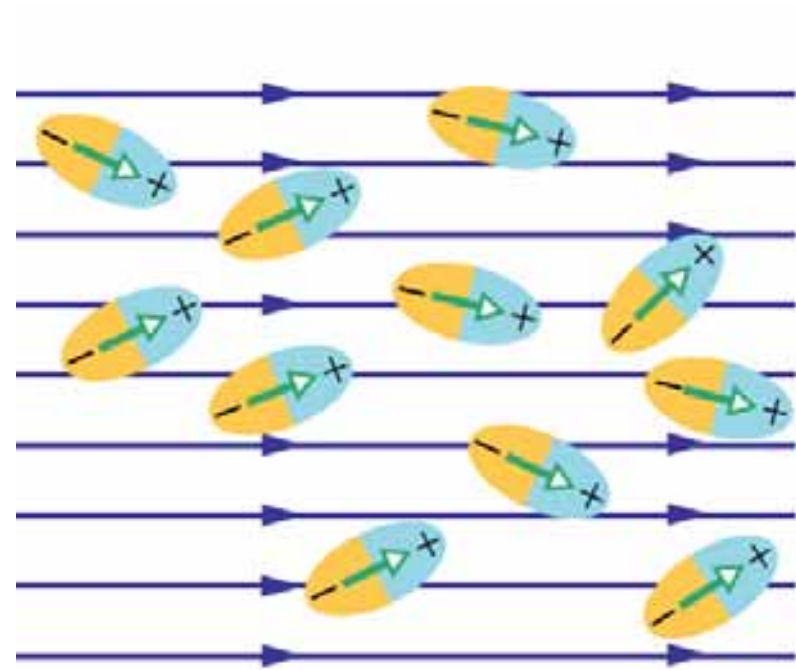
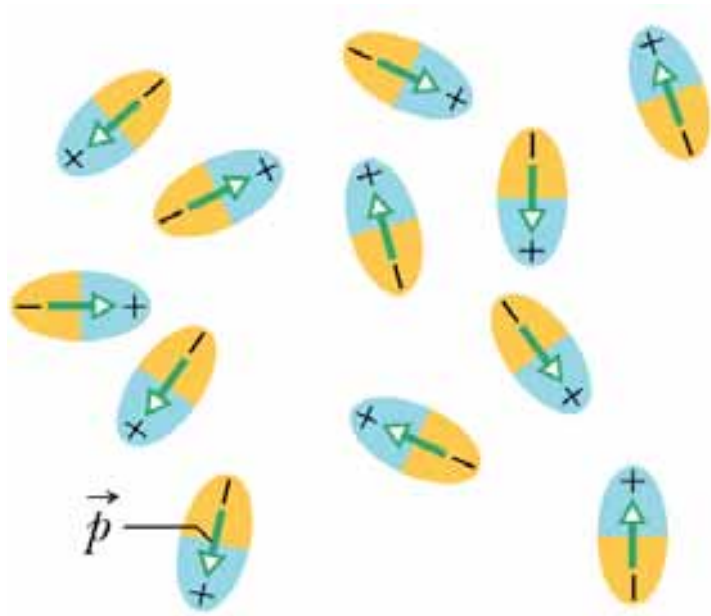
Halliday-Resnick-Walker (2003) Abb. 26.13

zu 3.3: Permanenter Dipol



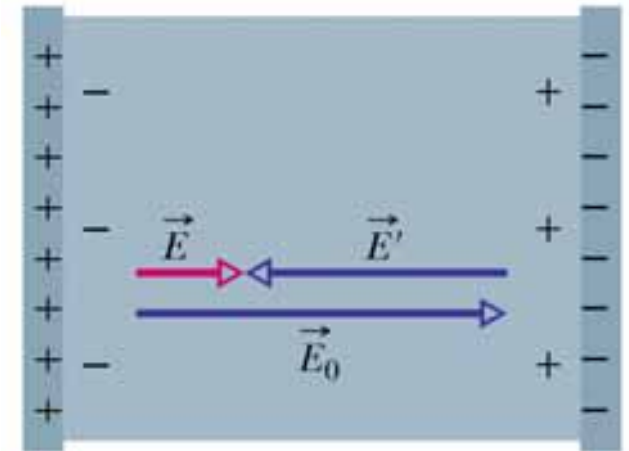
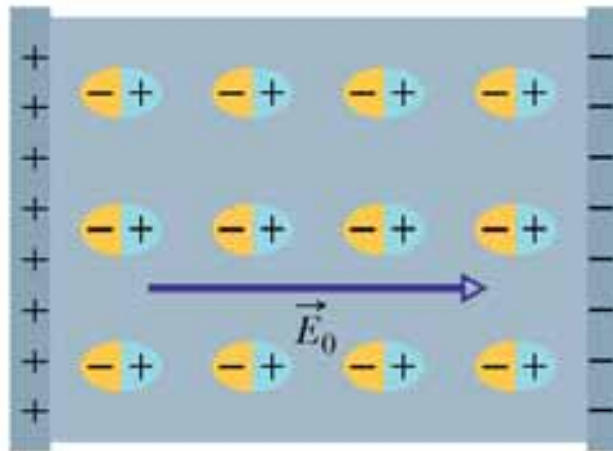
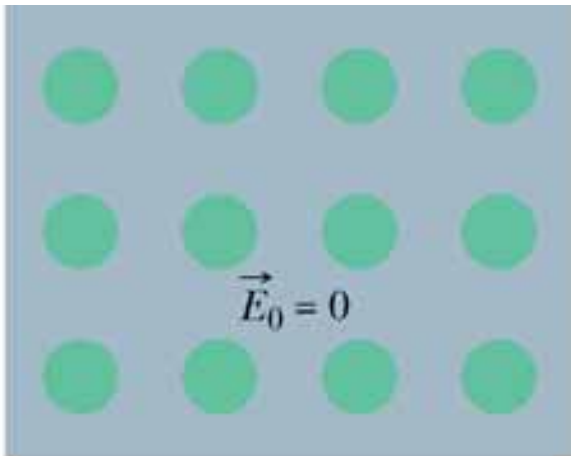
Halliday-Resnick-Walker (2003) Abb. 23.17

zu 3.3: Polares Dielektrikum



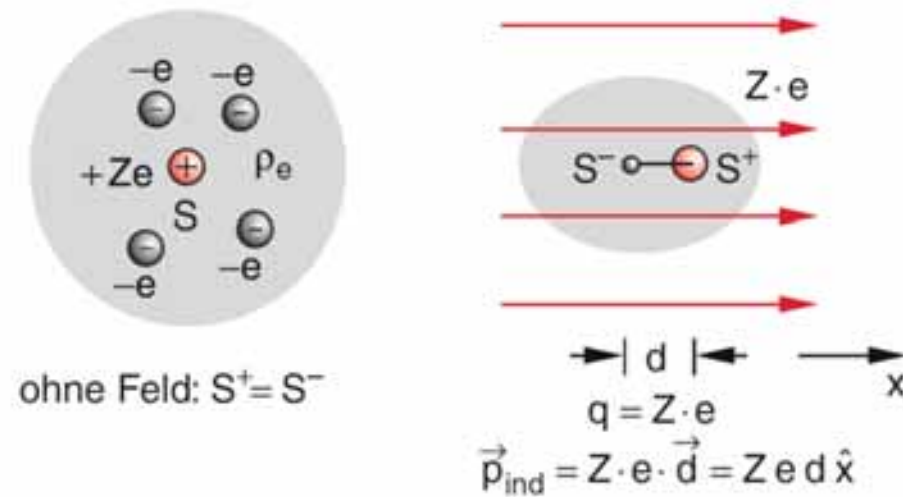
Halliday-Resnick-Walker (2003) Abb. 26.14

zu 3.3: Unpolares Dielektrikum



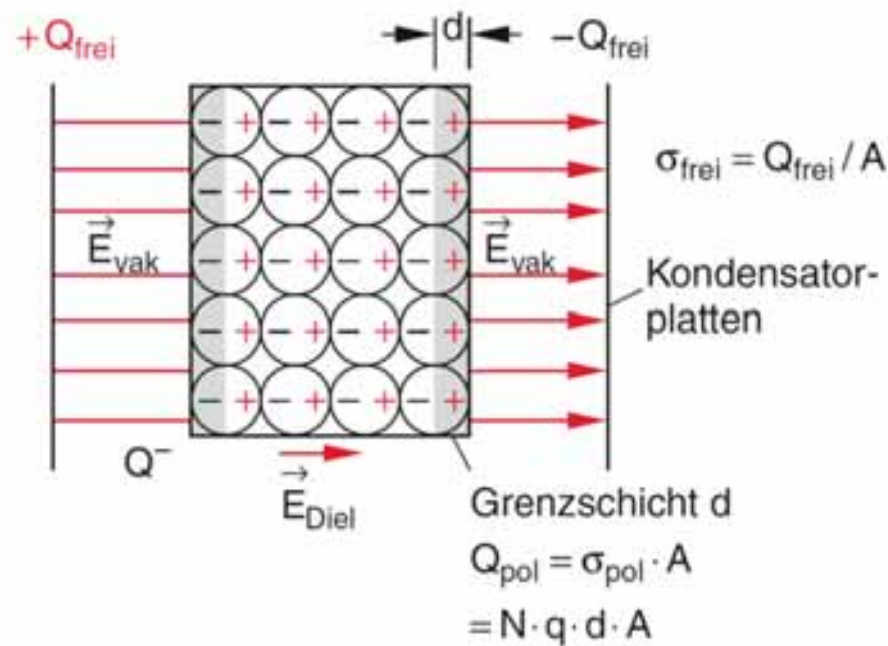
Halliday-Resnick-Walker (2003) Abb. 26.15

zu 3.3: Polarisation



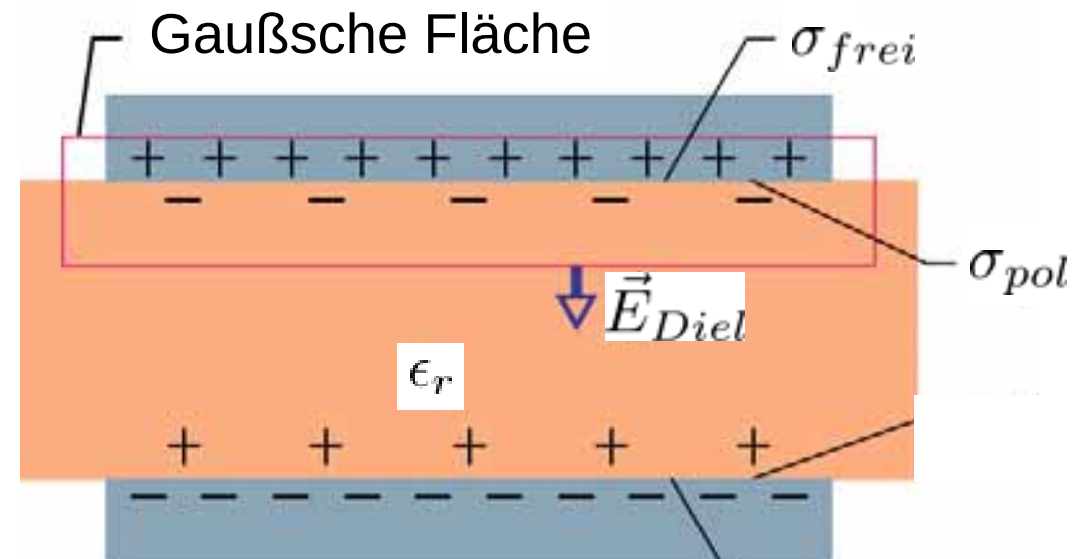
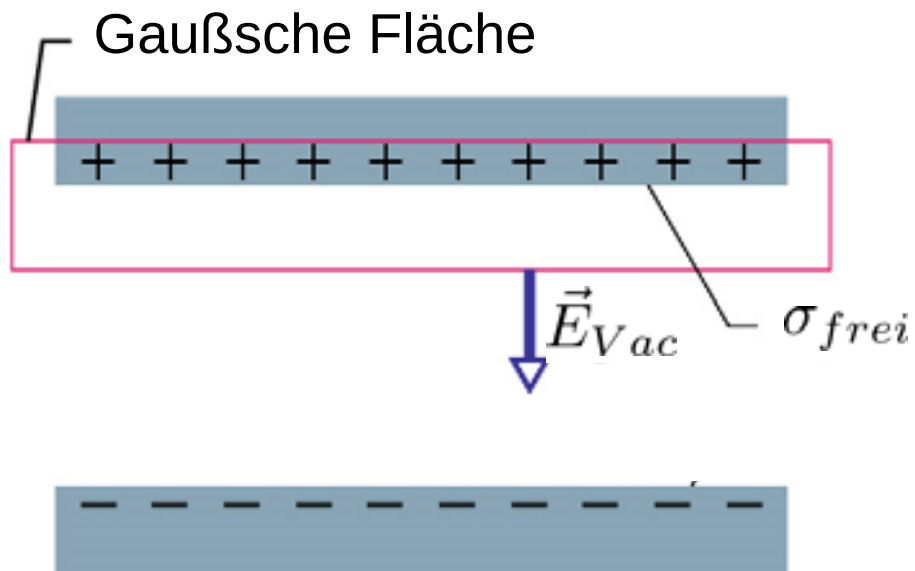
Demtröder – Experimentalphysik 2 (2013), Abb. 1.42.

zu 3.3: Polarisation



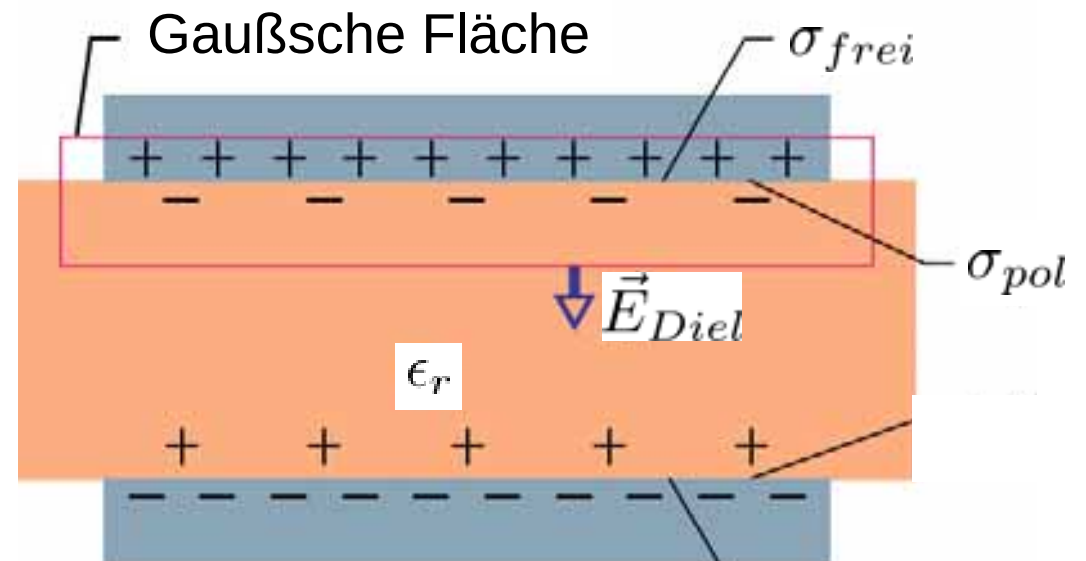
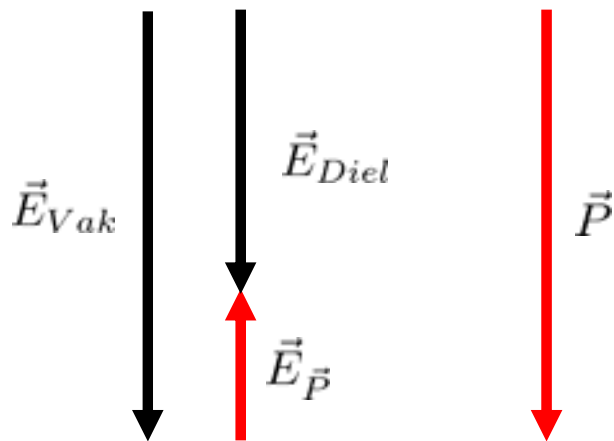
Demtröder – Experimentalphysik 2 (2013), Abb. 1.43.

zu 3.3: Gaußscher Satz im Dielektrikum



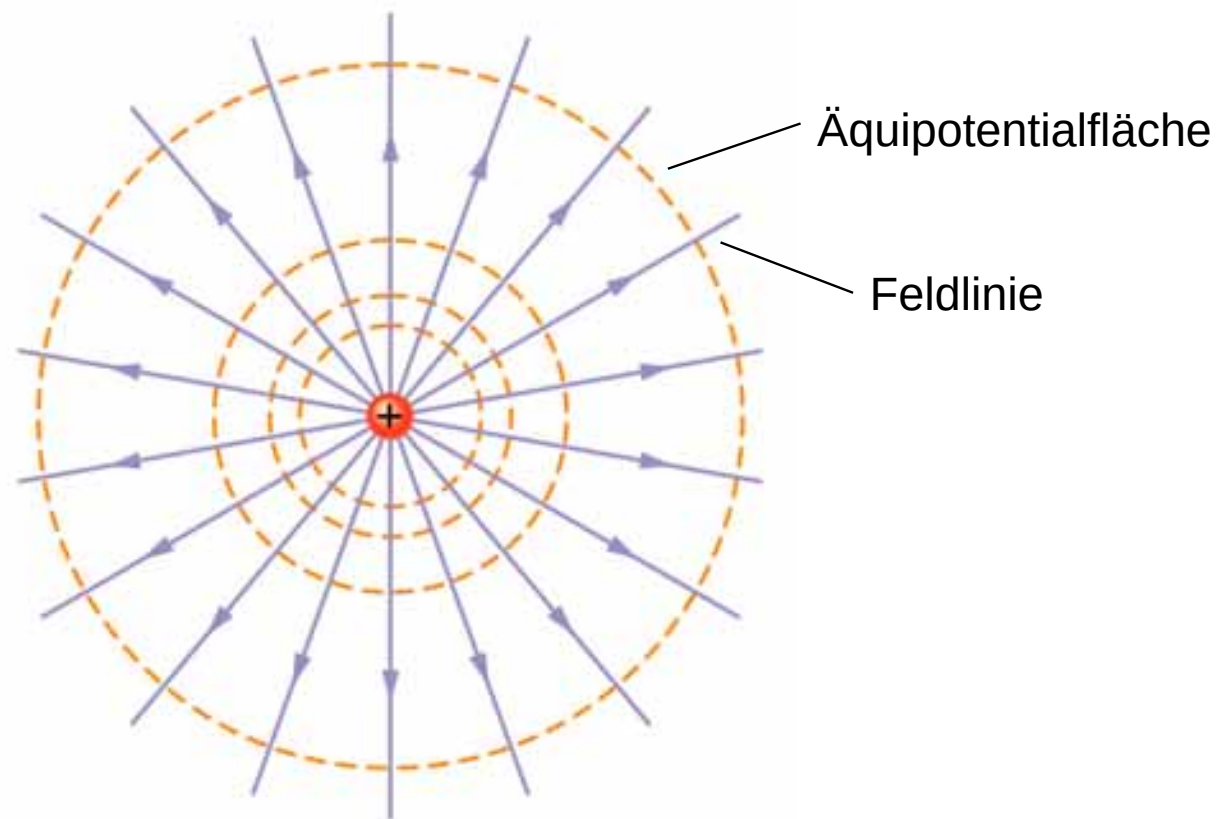
Halliday-Resnick-Walker (2003) Abb. 26.16

zu 3.3: Gaußscher Satz im Dielektrikum



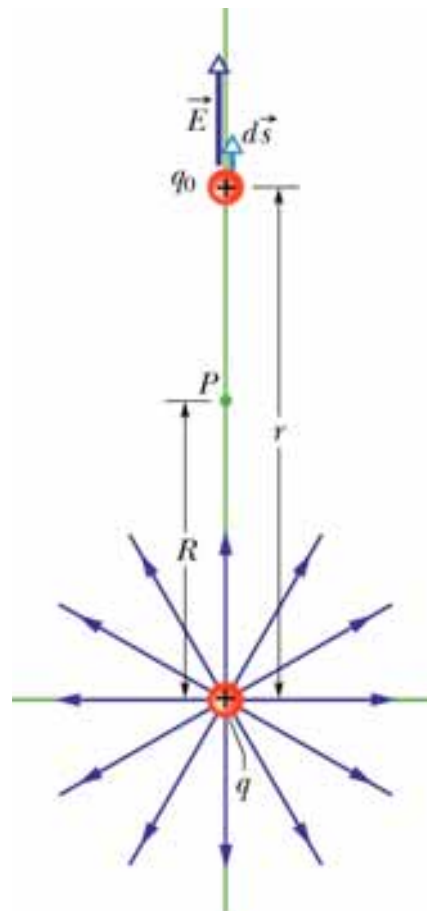
Halliday-Resnick-Walker (2003) Abb. 26.16

zu 3.4: Äquipotentialfläche einer Punktladung



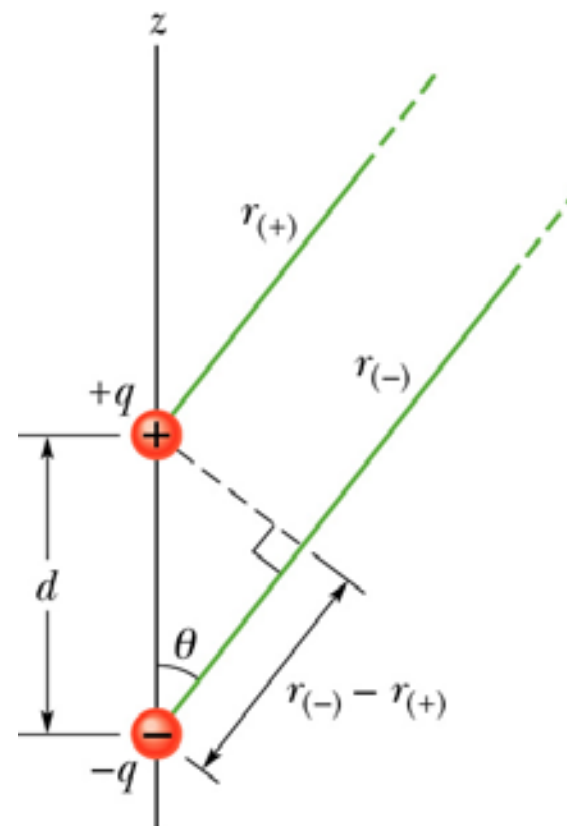
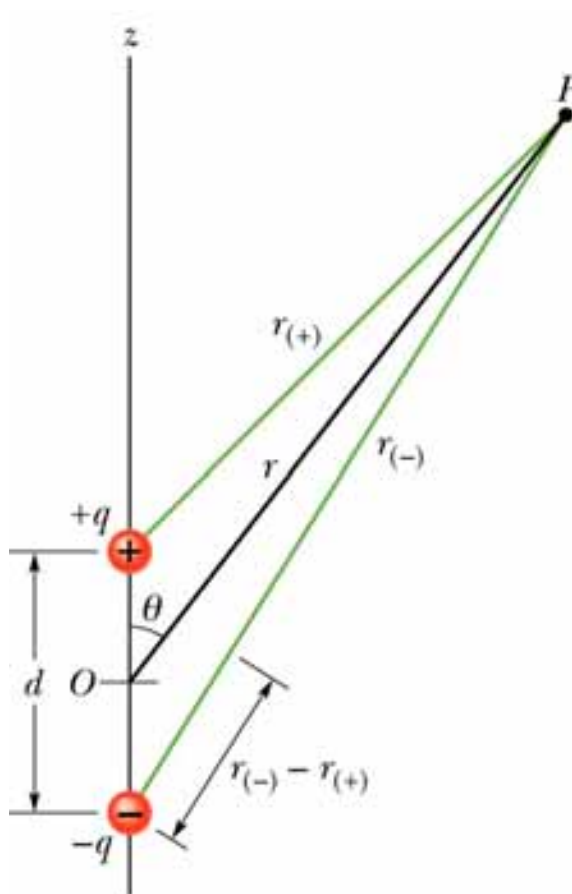
Halliday-Resnick-Walker (2003) Abb. 25.3

zu 3.4: Potential einer Punktladung



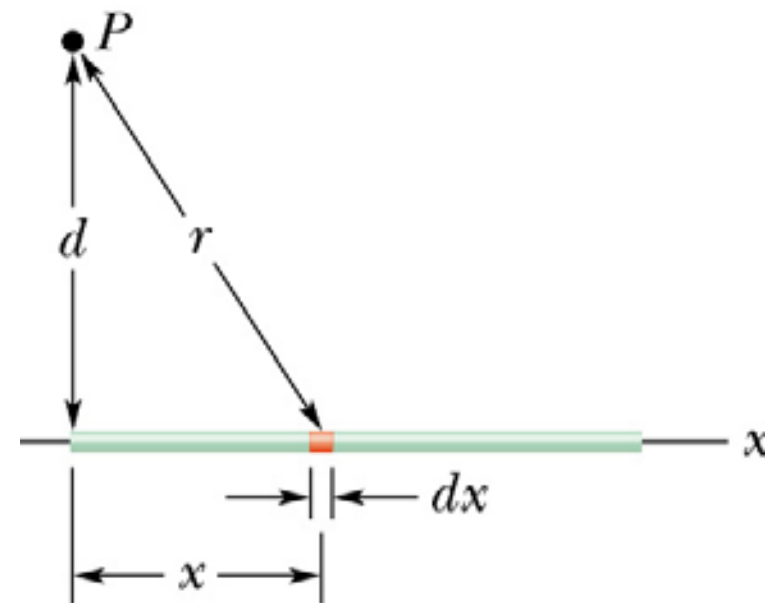
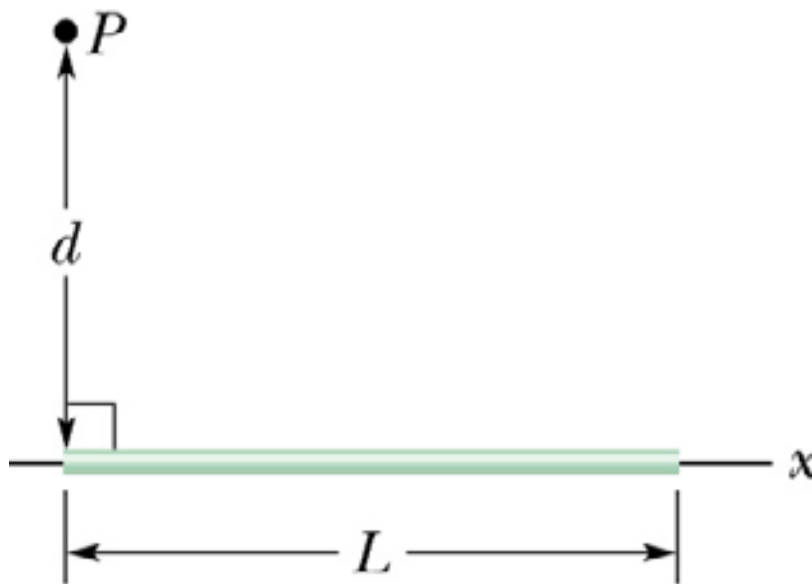
Halliday-Resnick-Walker (2003) Abb. 25.7

zu 3.4: Potential eines elektrischen Dipols



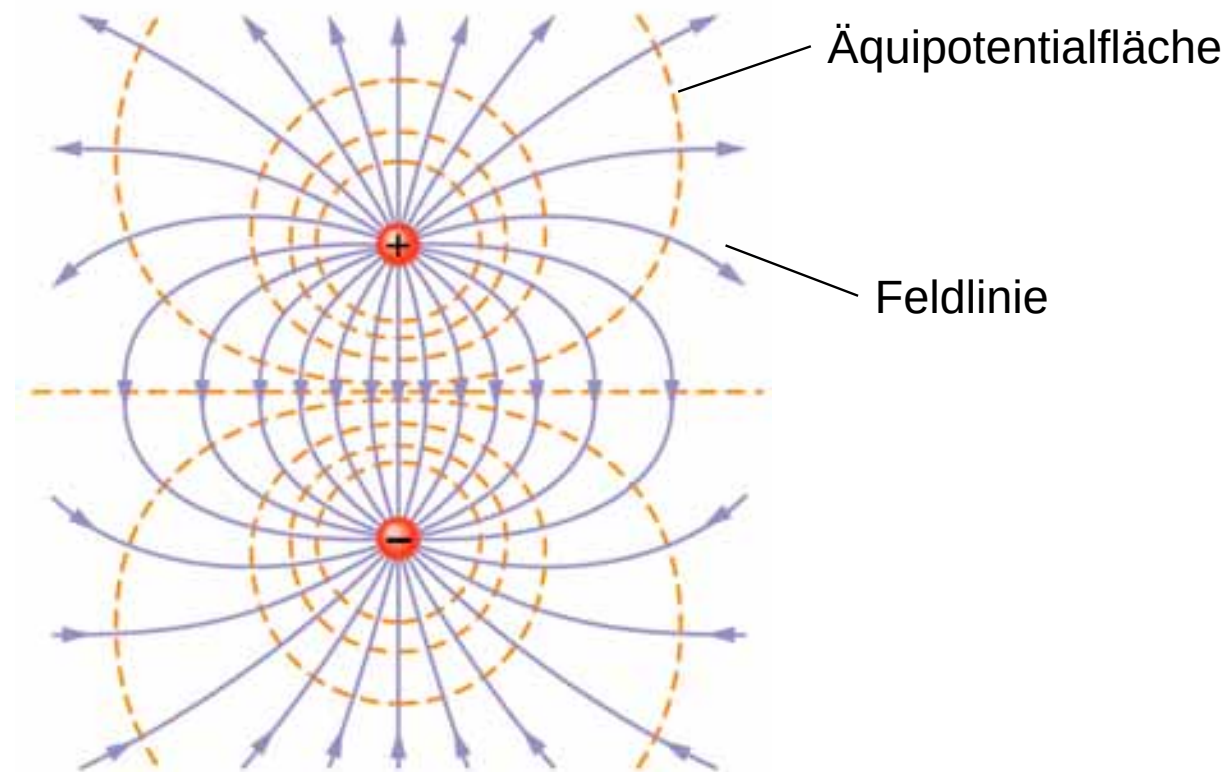
Halliday-Resnick-Walker (2003) Abb. 25.11

zu 3.4: Potential einer linearen Ladungsverteilung



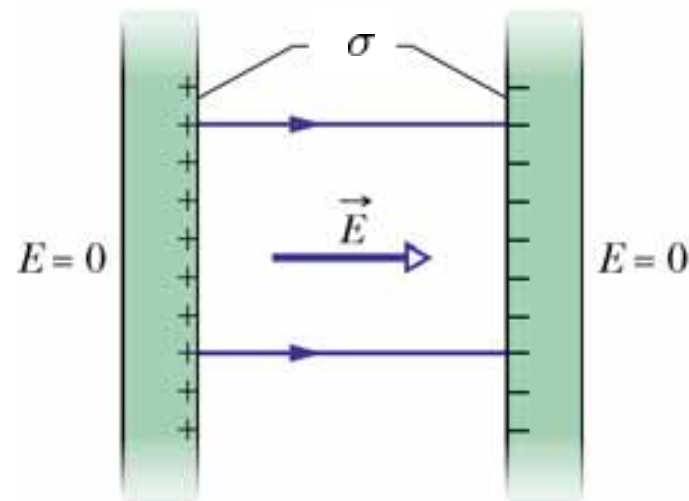
Halliday-Resnick-Walker (2003) Abb. 25.13

zu 3.4: Äquipotentialflächen eines elektrischen Dipols



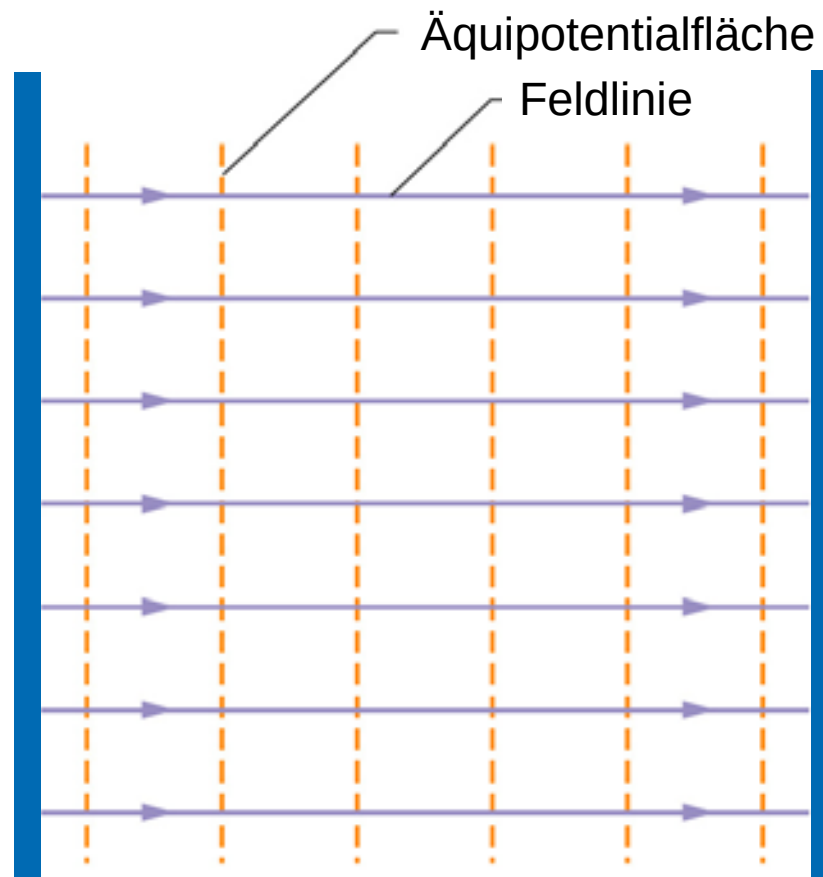
Halliday-Resnick-Walker (2003) Abb. 25.11

zu 3.4: Potential eines Plattenkondensators



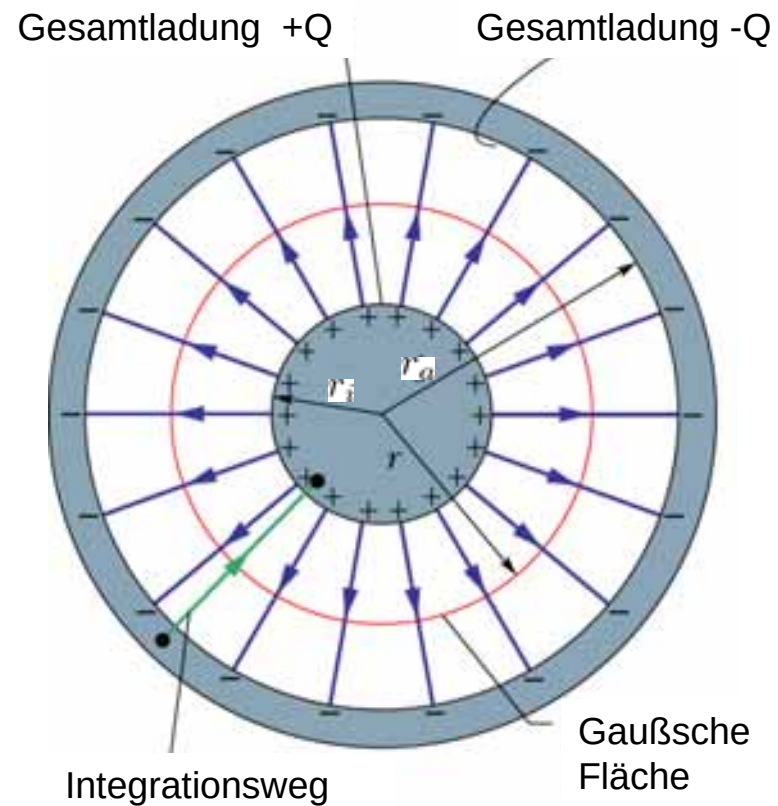
Halliday-Resnick-Walker (2003) Abb. 24.16

zu 3.4: Äquipotentialflächen eines Plattenkondensators



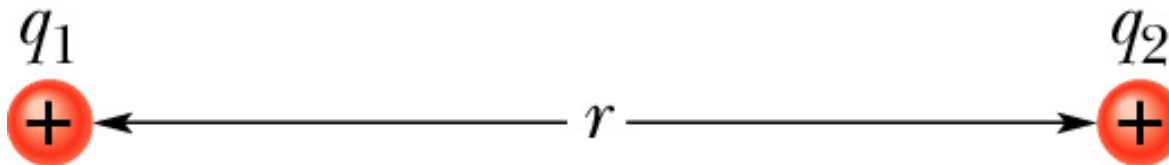
Halliday-Resnick-Walker (2003) Abb. 25.11

zu 3.4: Zylindekondensator



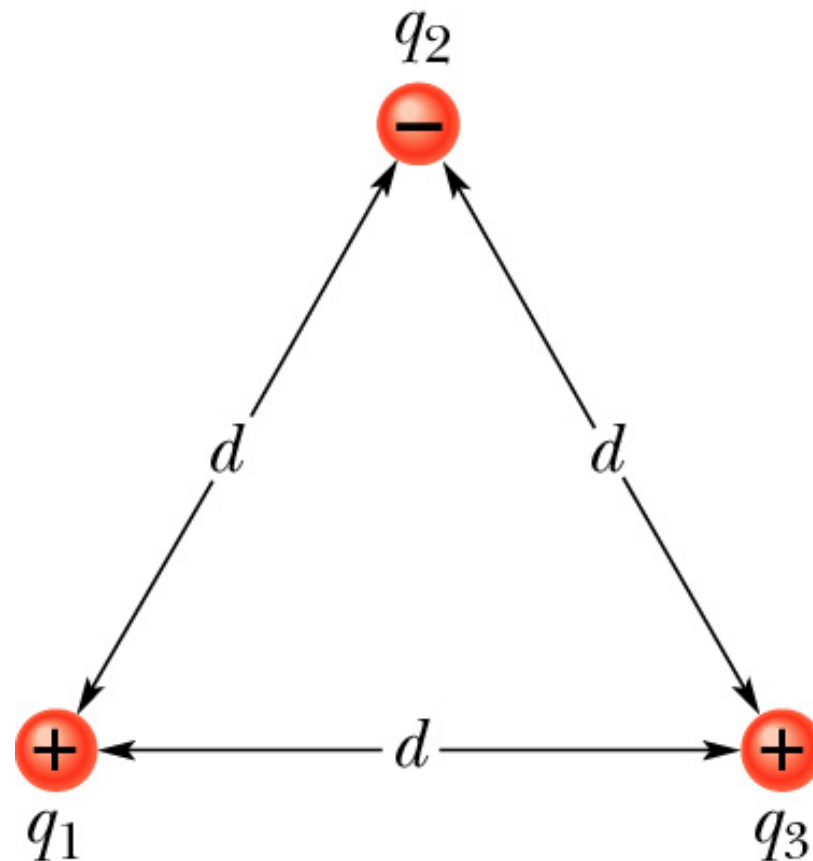
Halliday-Resnick-Walker (2003) Abb. 26.6

zu 3.4: Potentielle Energie in einem System von Ladungen



Halliday-Resnick-Walker (2003) Abb. 25.16

zu 3.4: Potentielle Energie in einem System von Ladungen



Halliday-Resnick-Walker (2003) Abb. 25.17